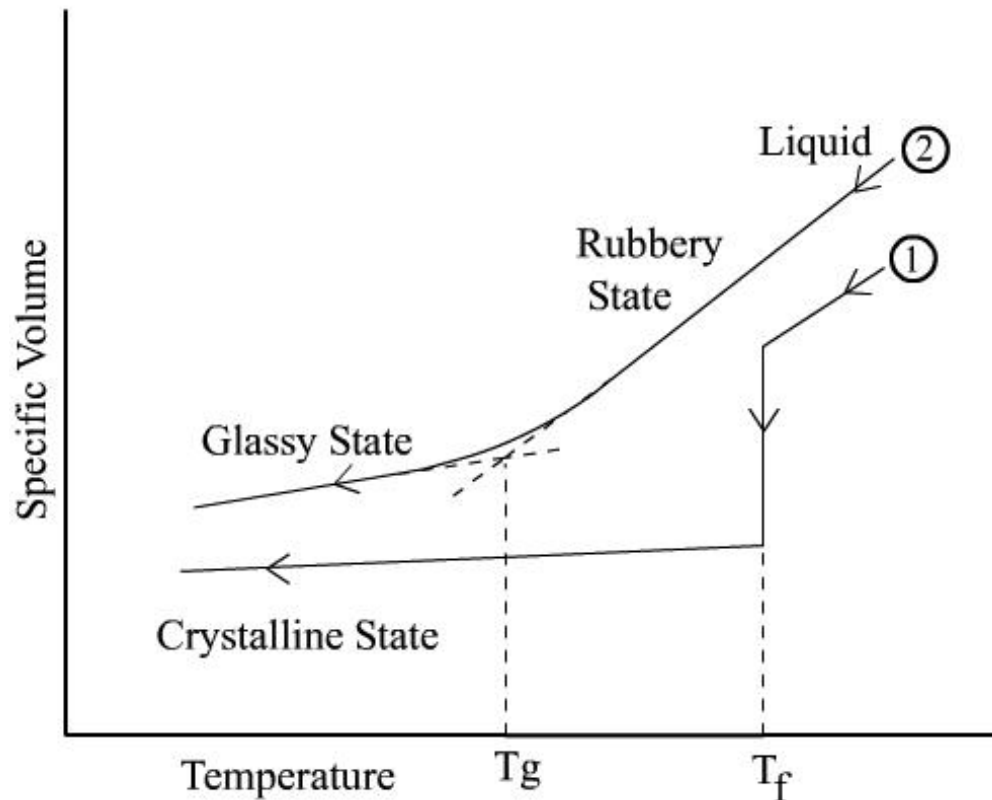


How to succeed

- Read the Chapter ahead of lectures
- Come to class
- Start paper early
- Study groups
- Prepare exams
- Don't cheat, plagiarize, or otherwise participate in un-ethical behavior
- **Learn with Internet: Wikipedia, google(baidu)...**
- Ask questions and Think skeptically

Important Points from Last Lecture

- Polymers are like snakes: reptation
- Glass Transition Temperature

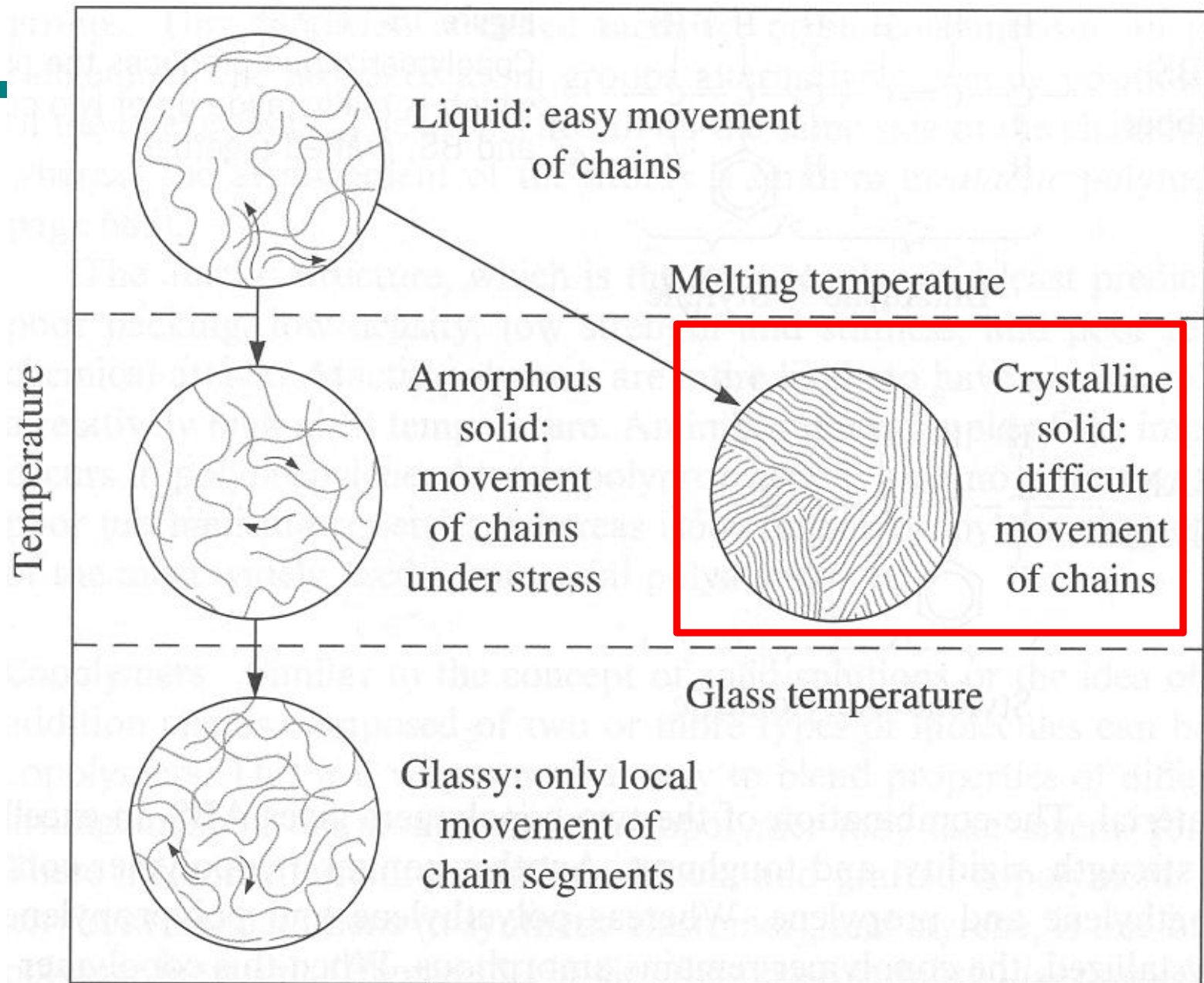




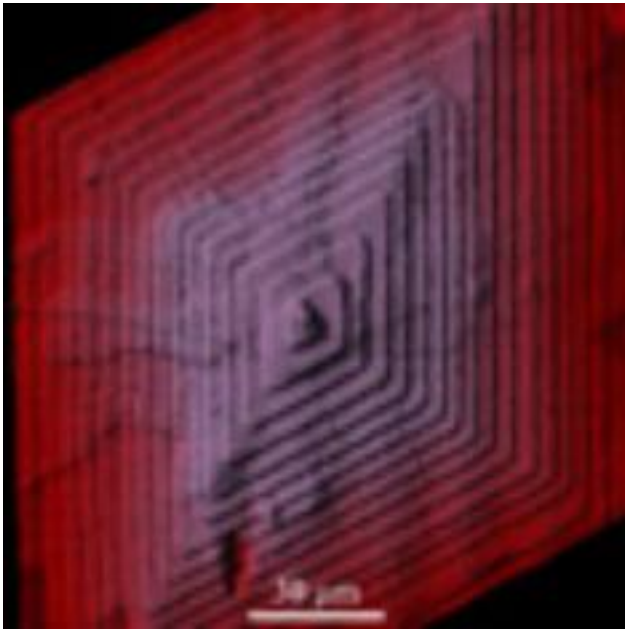
Introduction to Polymer Materials

Lecture 4-5: Solid State Properties II

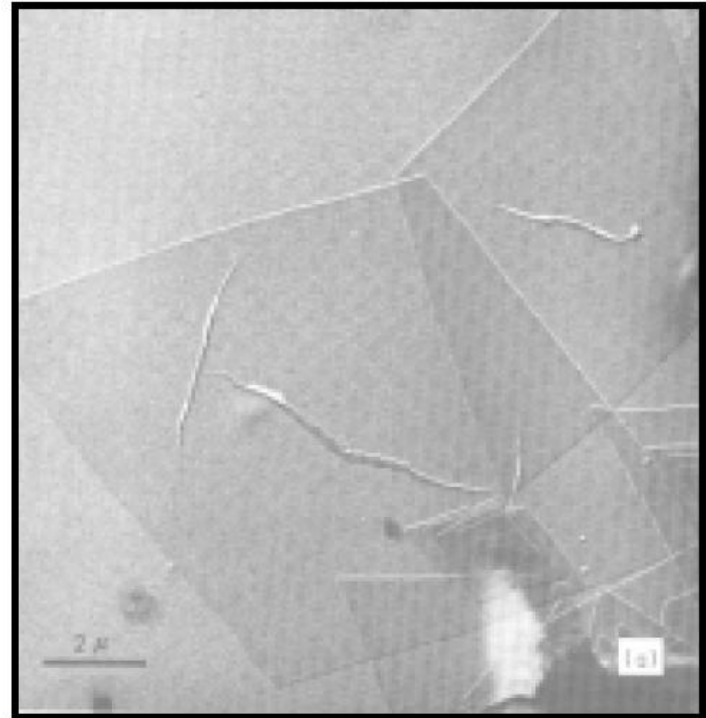
SUN Dazhi



Polymer Crystals

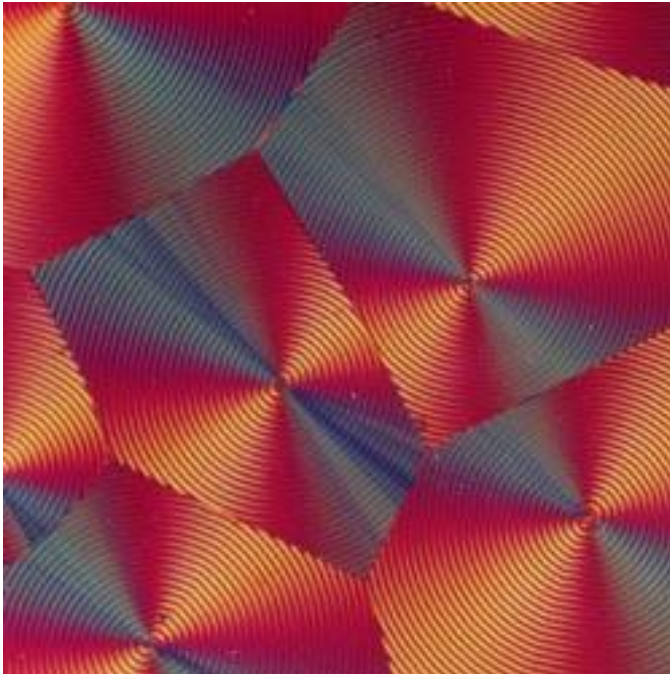


An image of an alkane crystal taken by AFM
System: Alkane, Composition: $C_{36}H_{74}$

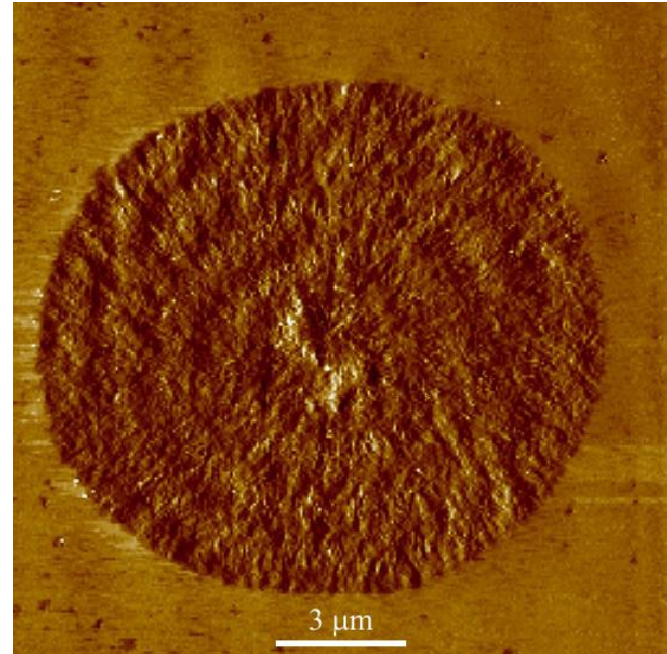


An image of a single crystal of polyethylene

Polymer Crystals

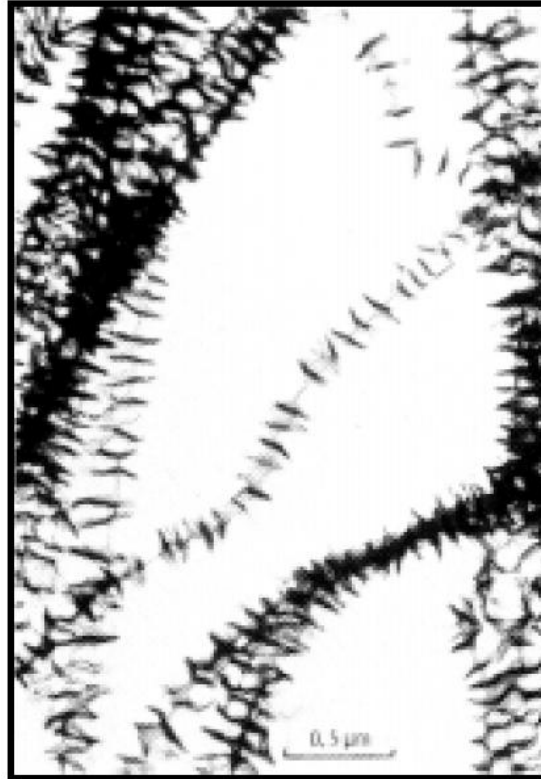


Maltese cross spherulites



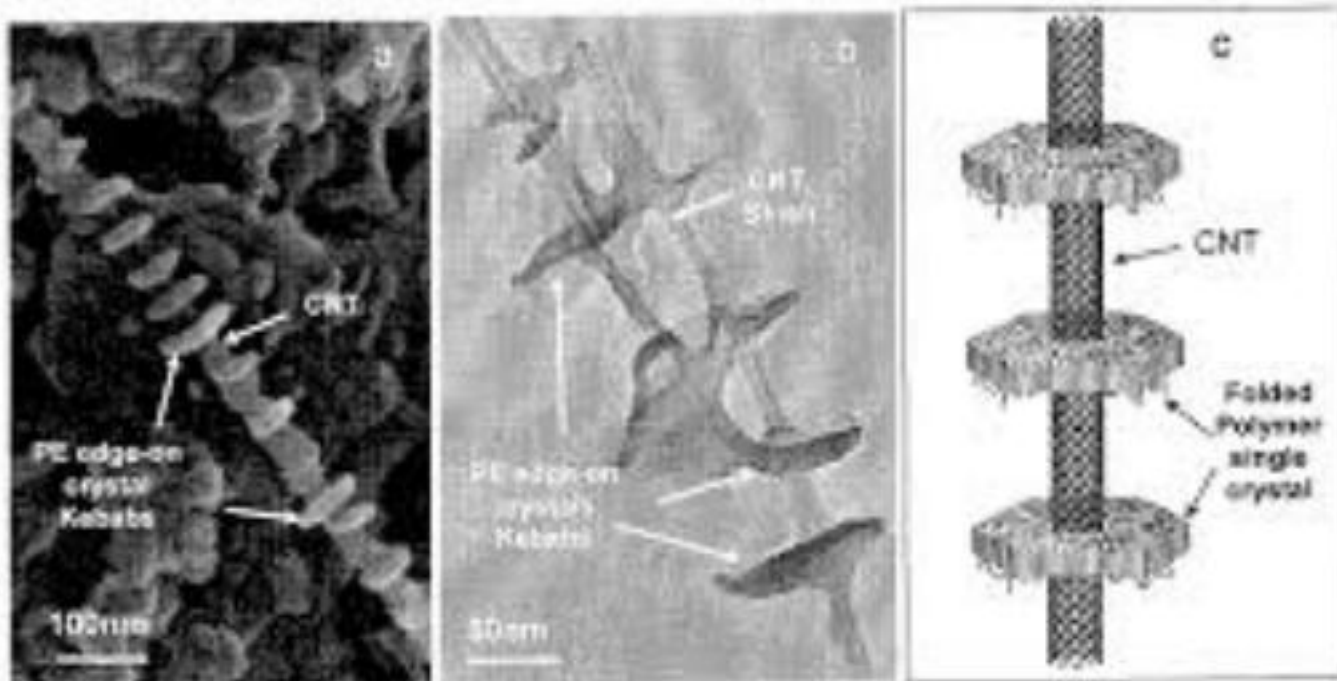
Single PE spherulite AFM

Polymer Crystals

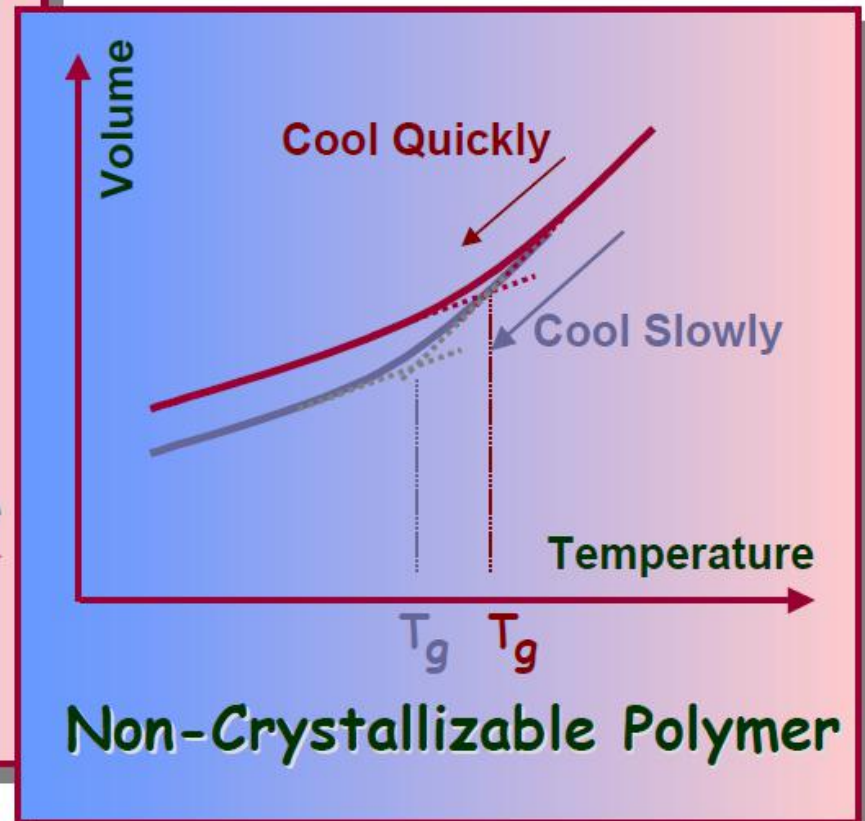
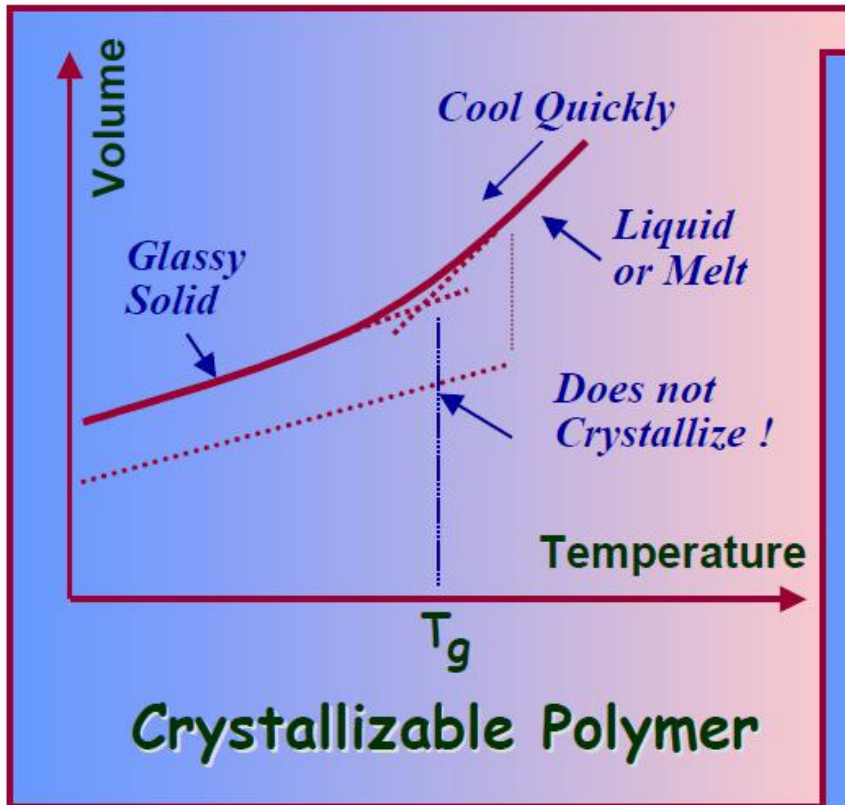


(a) Shish kebab morphology of polyethylene from solution (from Pennings, 1967³). (b) Shish kebabs of cellulose formed by recrystallizing cellulose II onto microfibrils of high molecular weight³.

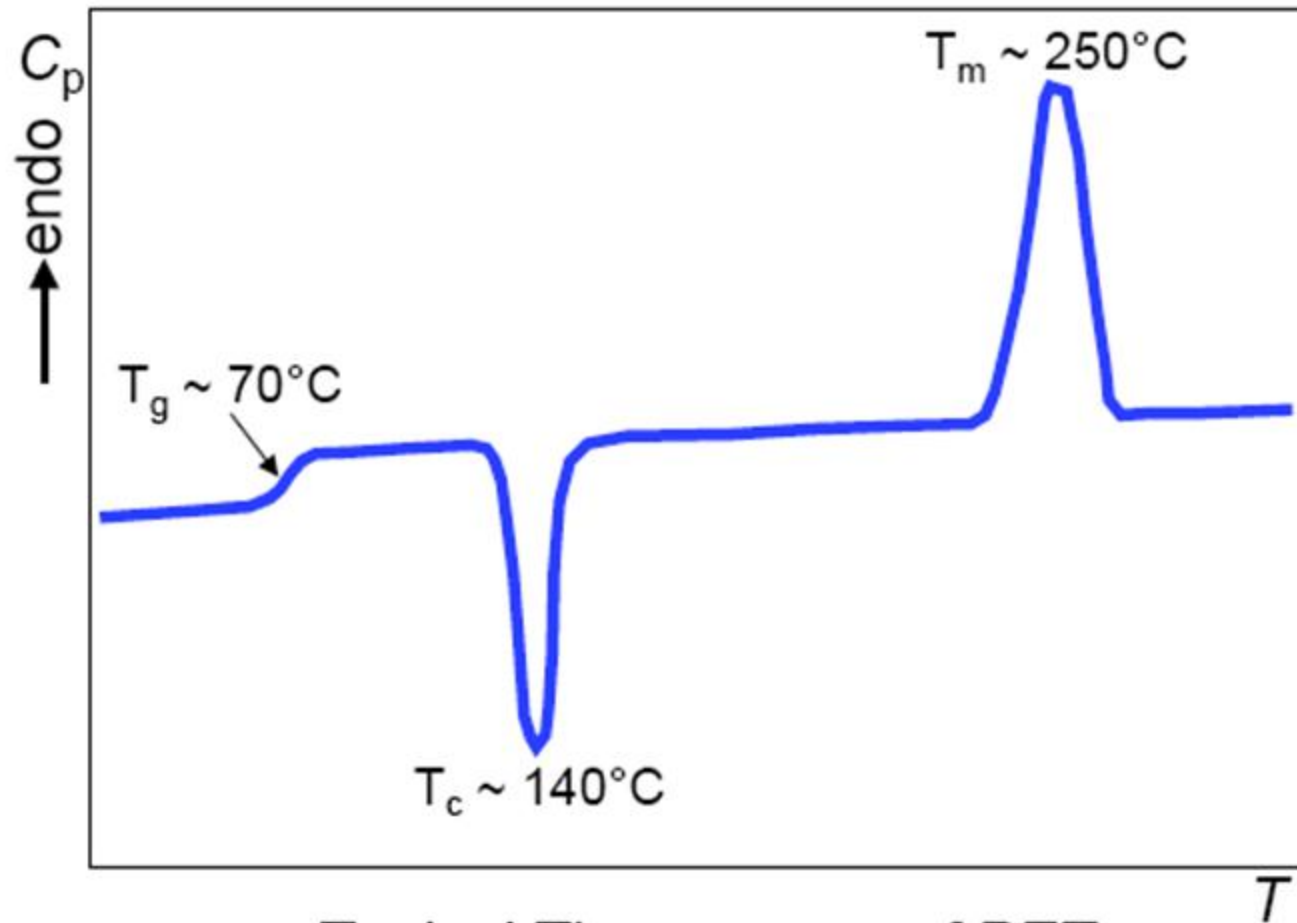
Polymer Crystals



Kinetics

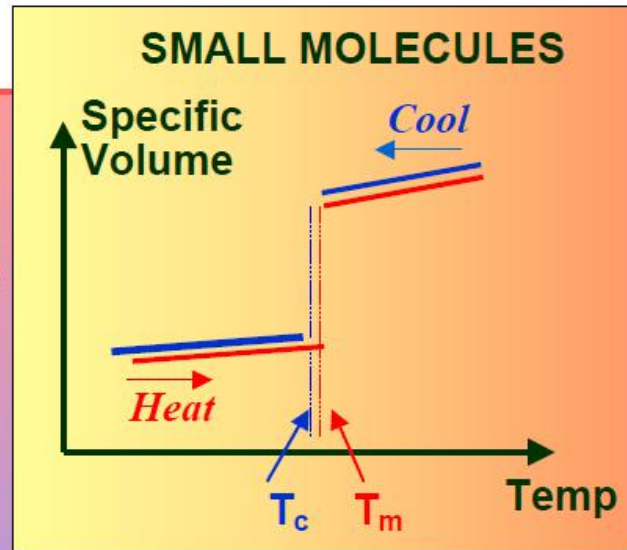
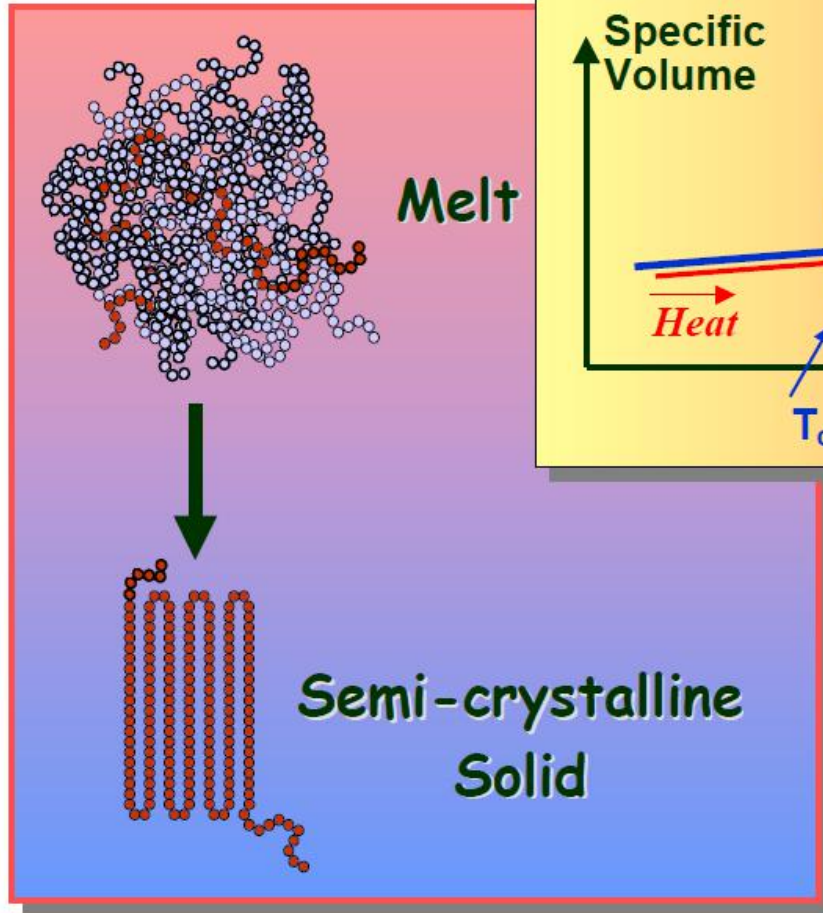


1. Crystallization and melting of crystalline polymers.
2. Glass transition of amorphous polymers.

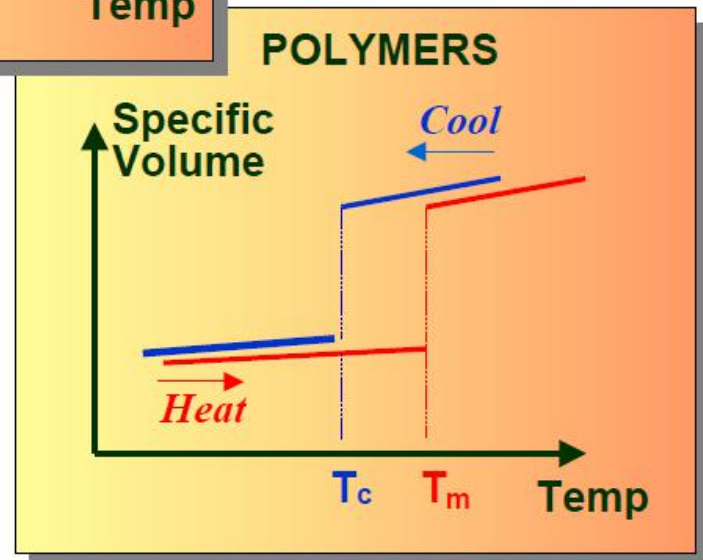


Typical Thermogram of PET

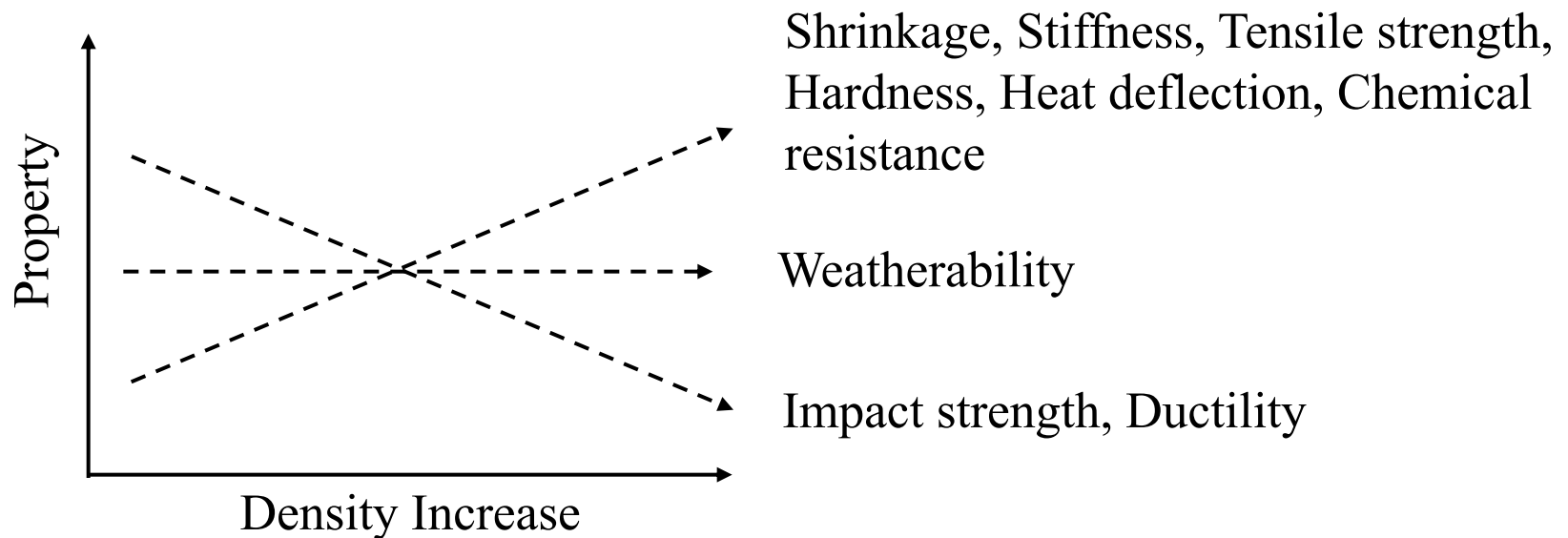
Polymer Crystallization



What is undercooling ?

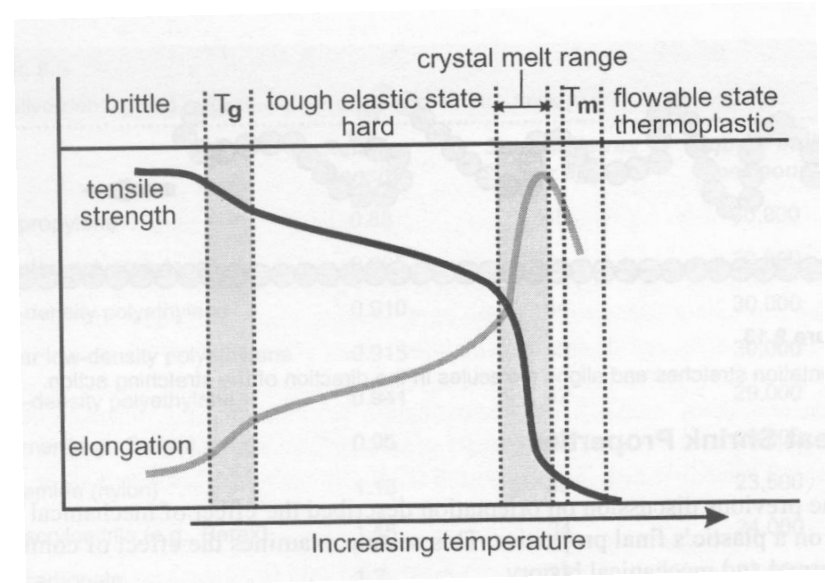
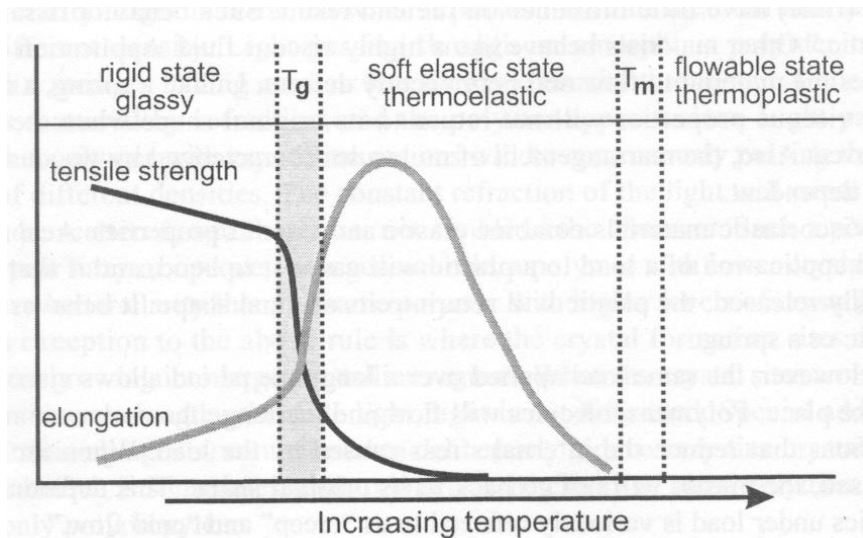


Material	Density (kg/m ³)	% Crystallinity
Low density polyethylene (LDPE)	915-929	45-65
Medium density polyethylene (MDPE)	930-939	65-75
High density polyethylene (HDPE)	940-965	75-90



Amorphous v Crystalline Polymers

Thermo-mechanical properties



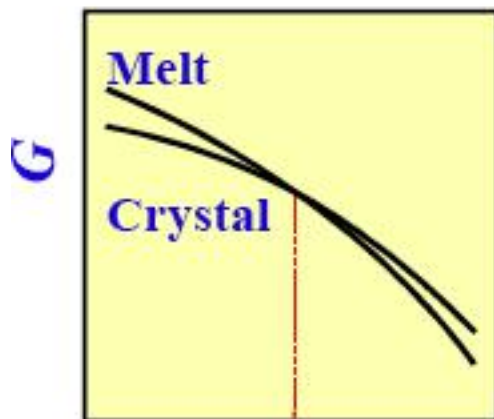
Thermodynamics of melting and crystallization

First order transitions

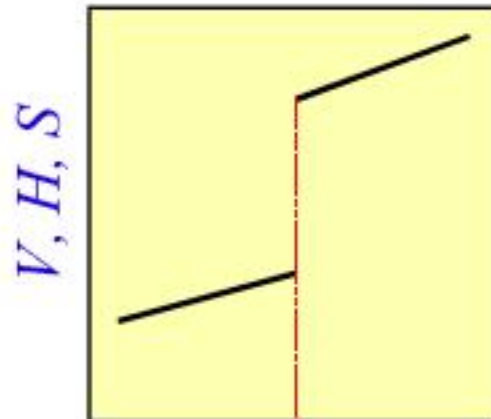
$$\Delta G_{\text{tran.}} = \Delta H_{\text{tran.}} - T_{\text{tran.}} \Delta S_{\text{tran.}} = 0 \quad \partial G / \partial T = -S$$

$$T_{\text{tran.}} = \Delta H_{\text{tran.}} / \Delta S_{\text{tran.}}$$

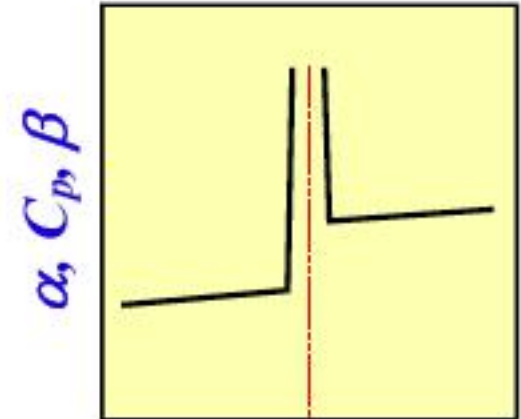
$$\partial^2 G / \partial T^2 = -C_p / T$$



$T_{\text{tran.}}$

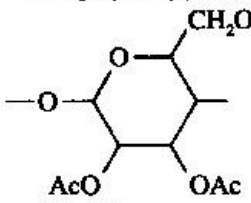


$T_{\text{trans.}}$



$T_{\text{trans.}}$

Thermal Transition Points of Select Polymers

Polymer	Repeating Unit	T_g (°C)	T_m (°C)
Polydimethylsiloxane	$-\text{OSi}(\text{CH}_3)_2-$	-127	-40
Polyethylene	$-\text{CH}_2\text{CH}_2-$	-125	137
Polyoxymethylene	$-\text{CH}_2\text{O}-$	-82	181
Natural rubber (polyisoprene)	$-\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_2-$	-73	28
Polyisobutylene	$-\text{CH}_2\text{C}(\text{CH}_3)_2-$	-73	44
Poly(ethylene oxide)	$-\text{CH}_2\text{CH}_2\text{O}-$	-41	66
Poly(vinylidene fluoride)	$-\text{CH}_2\text{CF}_2-$	-40	185
Polypropylene	$-\text{CH}_2\text{CH}(\text{CH}_3)-$	-8	176
Poly(vinyl fluoride)	$-\text{CH}_2\text{CHF}-$	41	200
Poly(vinylidene chloride)	$-\text{CH}_2\text{CCl}_2-$	-18	200
Poly(vinyl acetate)	$-\text{CH}_2\text{CH}(\text{OCOCH}_3)-$	32	
Poly(chlorotrifluoroethylene)	$-\text{CF}_2\text{CFCl}-$		220
Poly(ϵ -caprolactam)	$-(\text{CH}_2)_5\text{CONH}-$	52	223
Poly(hexamethylene adipamide)	$-\text{NH}(\text{CH}_2)_6\text{NHCO}(\text{CH}_2)_4\text{CO}-$	50	265
Poly(ethylene terephthalate)	$-\text{OCH}_2\text{CH}_2\text{OCO}-$  $-\text{CO}-$	61	270
Poly(vinyl chloride)	$-\text{CH}_2\text{CHCl}-$	81	273
Polystyrene	$-\text{CH}_2\text{CH}\phi-$	100	250
Poly(methyl methacrylate)	$-\text{CH}_2\text{C}(\text{CH}_3)(\text{CO}_2\text{CH}_3)-$	105	200
Cellulose triacetate			306
Polytetrafluoroethylene	$-\text{CF}_2\text{CF}_2-$	117	327

Rule of Thumb for T_g 's and T_m 's

- For symmetrical polymers: $T_g = 0.5 T_m$ (Kelvin)

Polyvinylidene chloride $T_g = -18 + 273 = 255 \text{ K}$

$$T_m = T_g / 0.50 = 255 / 0.5 = 510 \text{ K or } 237^\circ \text{ C}$$

Experimentally $T_m = 200^\circ \text{ C}$

- For asymmetrical polymers: $T_g = 0.66 T_m$ (Kelvin)

Polyvinyl chloride $T_g = 81 + 273 = 354 \text{ K}$

$$T_m = T_g / 0.66 = 354 / 0.66 = 536 \text{ K or } 263^\circ \text{ C}$$

Experimentally $T_m = 273^\circ \text{ C}$

Rule of Thumb for T_g 's and T_m 's

Caution: Its just a rule of thumb:

Atactic polystyrene $T_g = 104 + 273 = 377$ K

$T_m = T_g/0.66 = 377/0.66 = 571$ K or 298° C

Experimentally $T_m = 523$ K or 250° C

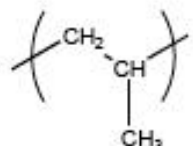
Crystalline Polymers (really semicrystalline)

Basic Requirement: Chain Regularity to Allow Close Packing

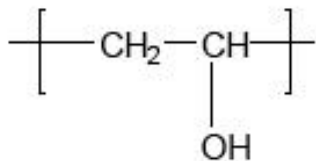
Example Crystalline Polymers:

1. Polyethylene.
2. Nylon 6,6.
3. Poly(ethylene terephthalate)

Vinyl Polymers with Stereoregularity

Isotactic or syndio-tactic polypropylene 

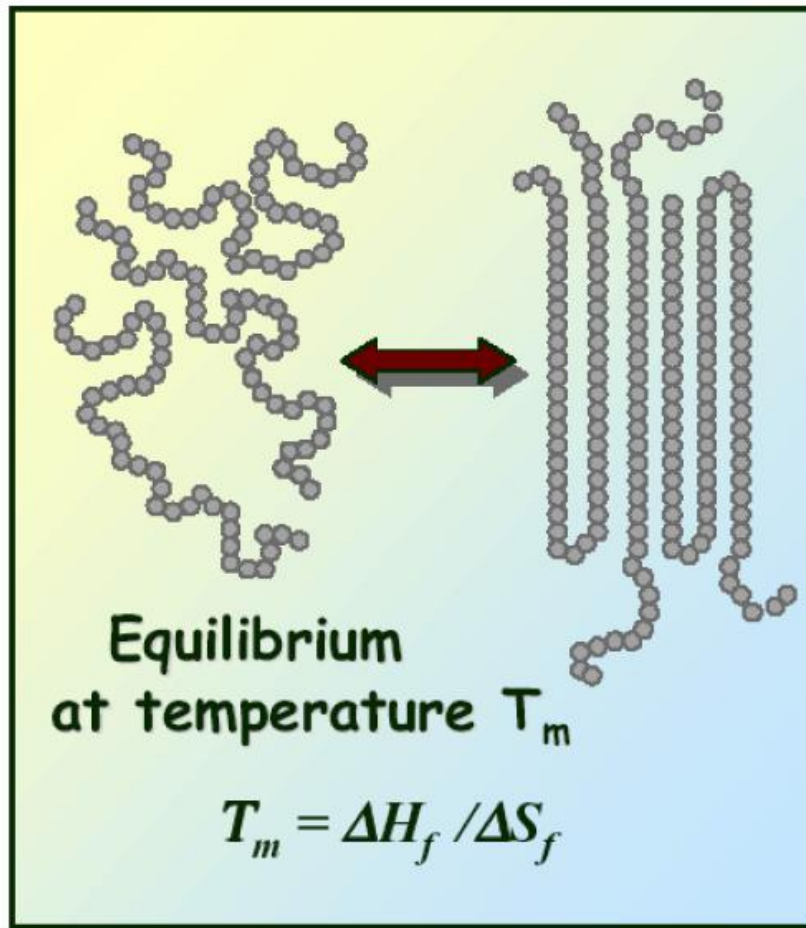
Exception: Atactic polyvinyl alcohol



Why?

-OH is sufficiently similar in size that they can fit together well enough.

Thermodynamic of Crystallization



$$\Delta G_f = \Delta H_f - T\Delta S_f$$

And at Equilibrium

$$\Delta G_f = 0$$

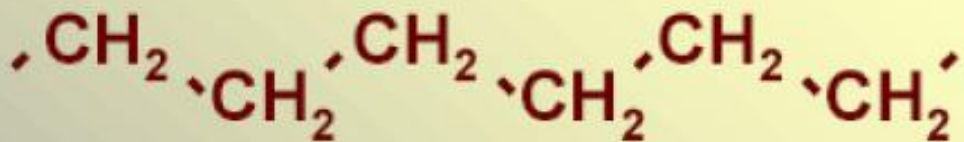
Hence

$$T_m = \Delta H_f / \Delta S_f$$

The subscript “f”
stands for fusion

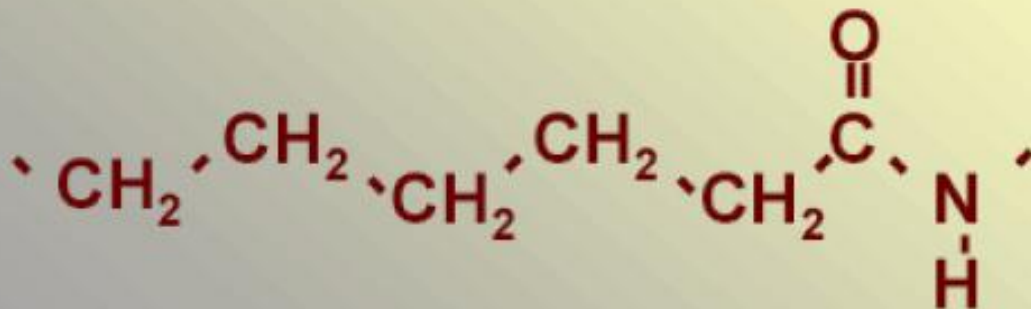
For melting ΔS_f is positive

Intramolecular interactions (ΔH_f) favor crystallization & higher T_m



$$T_m \sim 135^\circ\text{C}$$

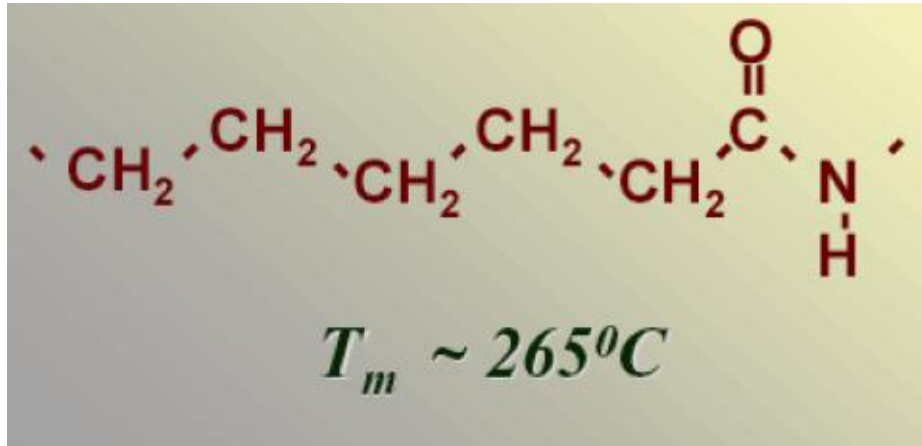
Van der Waals:
2 kJ/mole



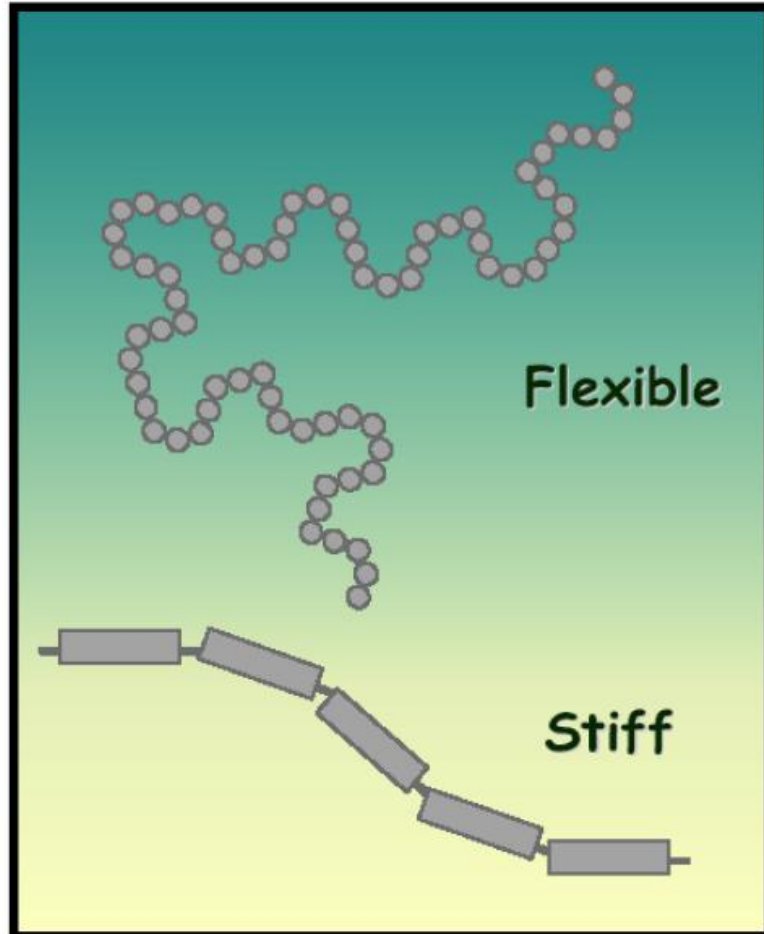
$$T_m \sim 265^\circ\text{C}$$

Hydrogen bonding
20 kJ/mol

Explain why Nylon 6 has a lower T_m than Kevlar

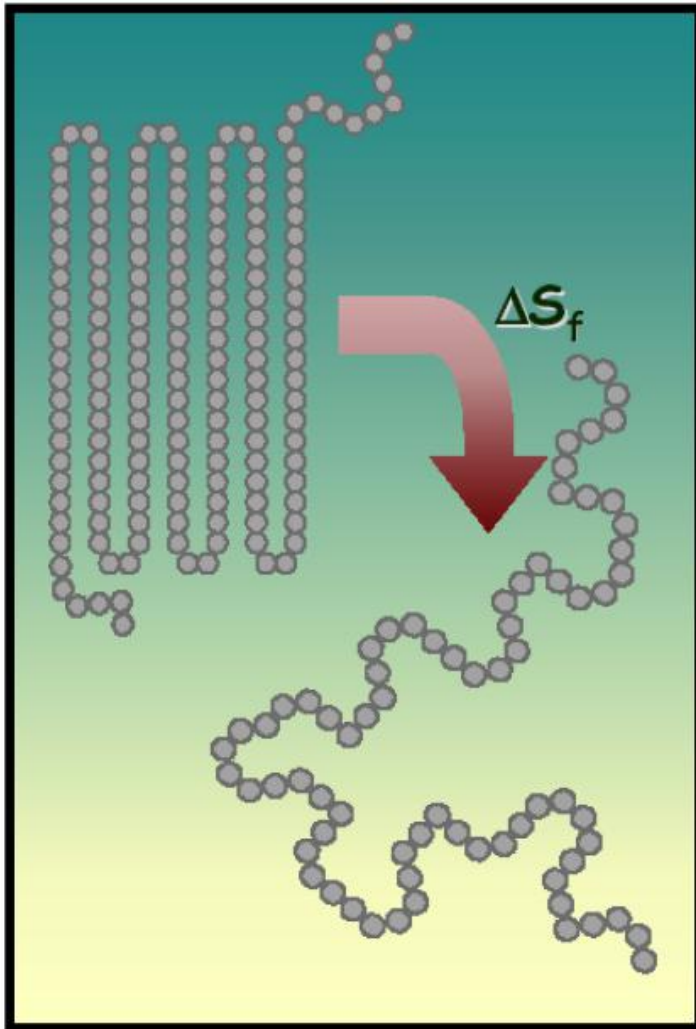


Entropic Contributions to T_m



$$T_m = \Delta H_f / \Delta S_f$$

Flexible Chains have numerous conformations



$$S = k \ln \Omega$$

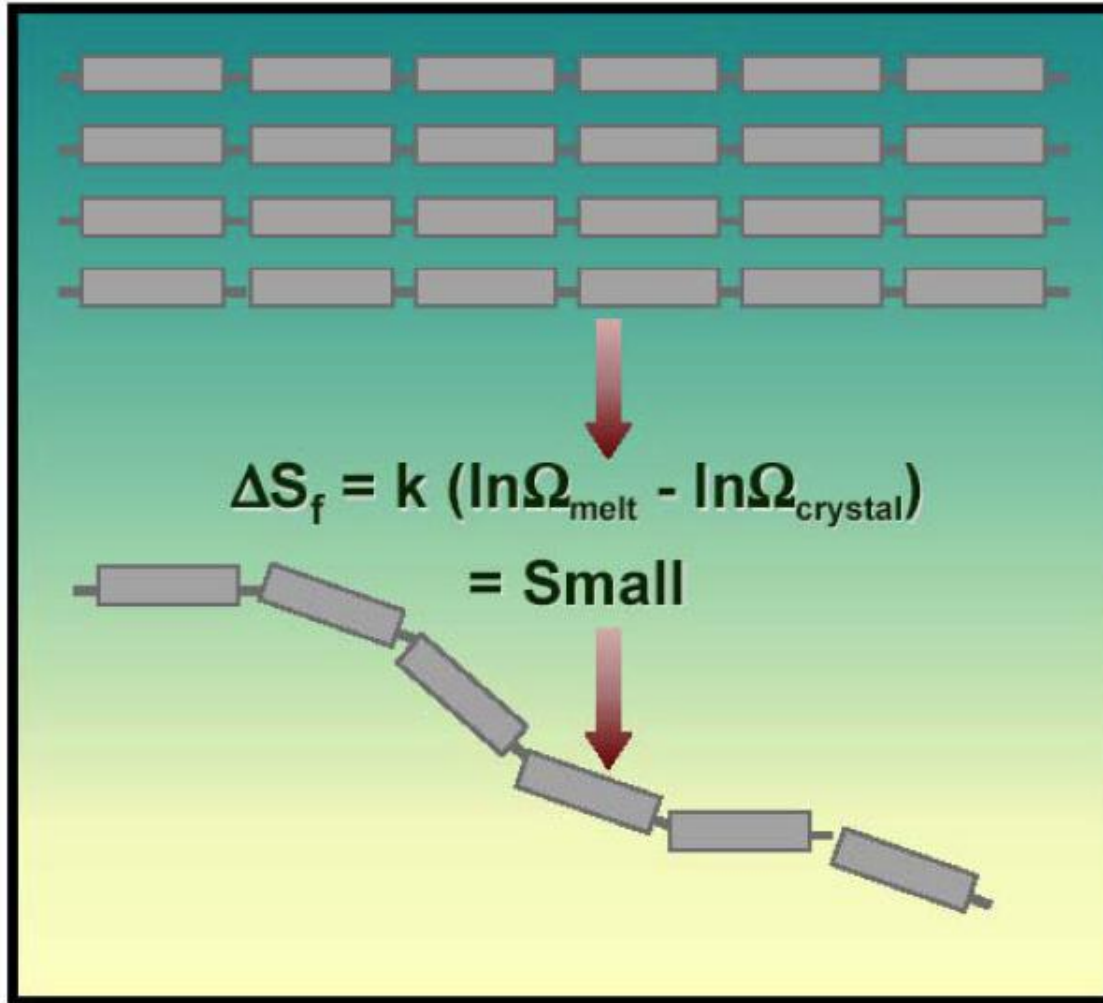
$$\Delta S_f = k (\ln \Omega_{\text{melt}} - \ln \Omega_{\text{crystal}}) = \text{Large}$$

ΔS_f : Large

T_m : Low

Nylon 6

Rigid Chains have fewer conformations

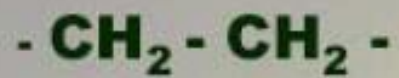


$$T_m = \Delta H_f / \Delta S_f$$

ΔS_f : Small

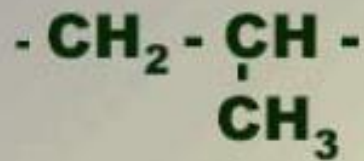
T_m : High

Kevlar example



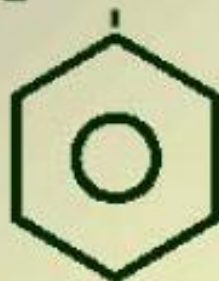
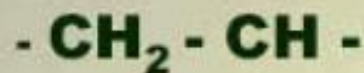
Polyethylene

$$T_m \sim 135^{\circ}\text{C}$$



Isotactic Polypropylene

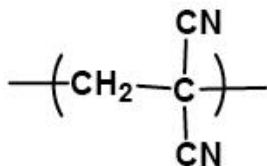
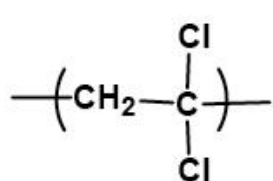
$$T_m \sim ?$$



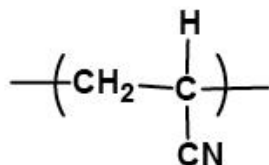
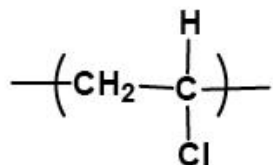
Isotactic Polystyrene

$$T_m \sim ?$$

Polymer symmetry and Melting Point

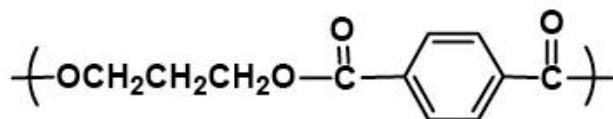


Highly crystalline

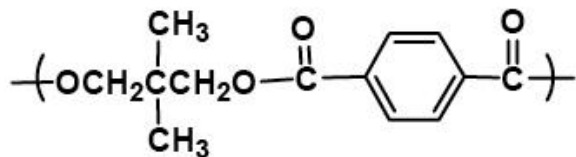


Atactic

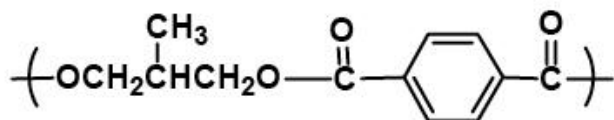
Very low degree of crystallinity



Crystalline m.p. 220 °C

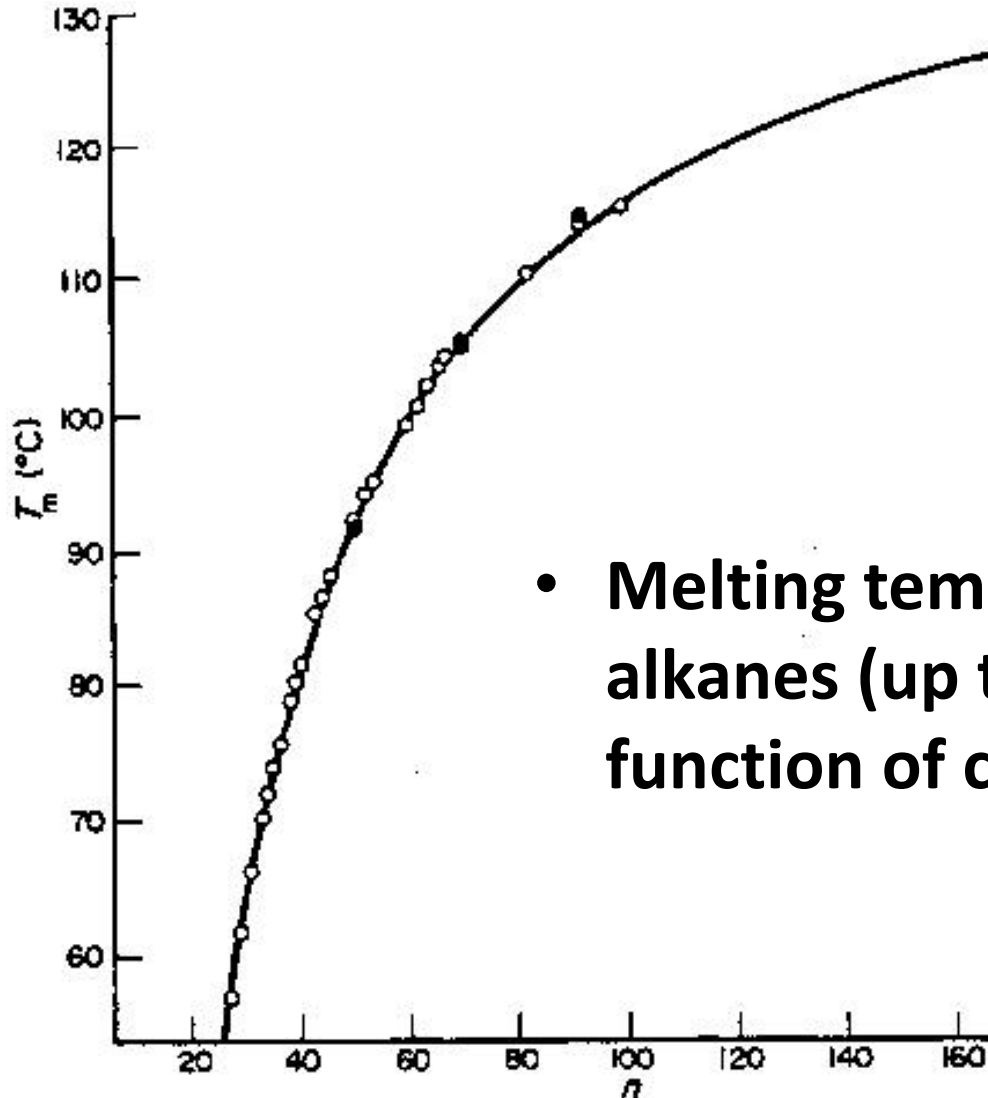


Crystalline m.p. 140 °C



Noncrystalline

Molecular Weight Influence on T_m



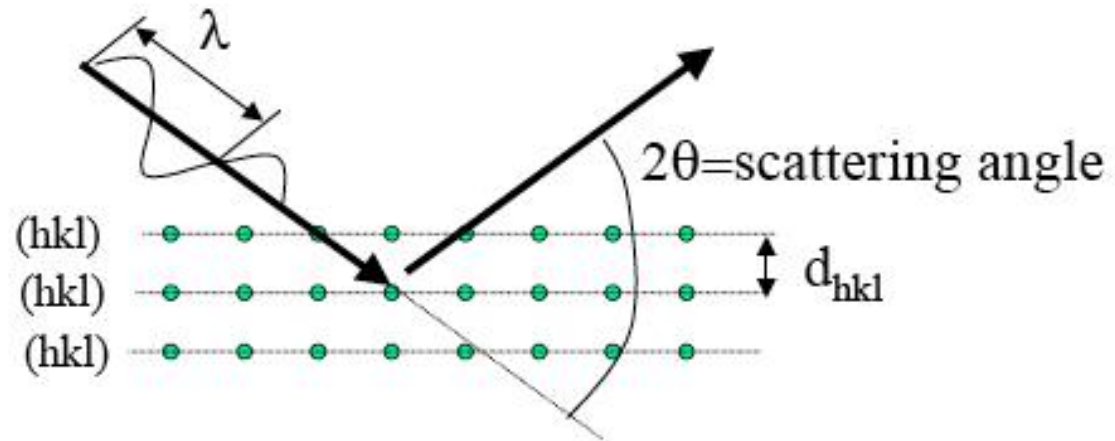
- Melting temperatures of n-alkanes (up to C_{100}) as a function of chain length.

Methods for Inducing Crystallization in Polymers

- Slow cooling of molten polymer
- Annealing between T_g and T_m
- Evaporation of solvent
- Shear & disentanglement
- Stretching and alignment of macromolecules

Characterization of Crystalline Polymers: Diffraction

Bragg's Law

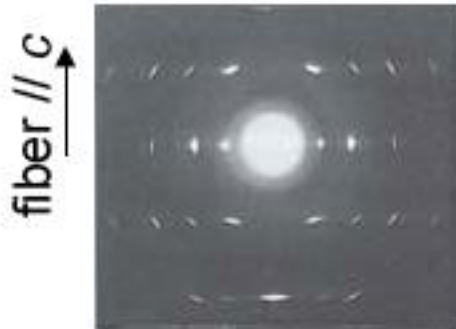


The condition for *diffraction* of x-rays

$$n\lambda = 2d_{(hkl)} \sin \theta$$

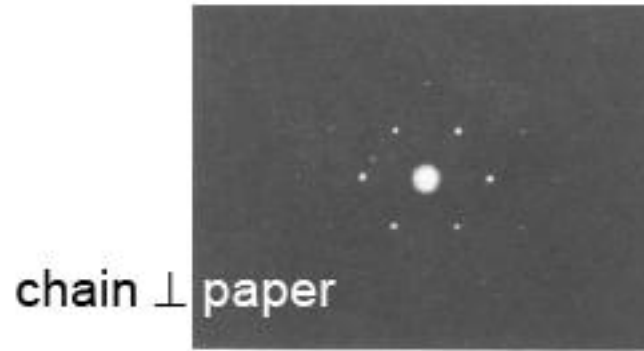
integer → n
x-ray wavelength → λ
lattice spacing → $d_{(hkl)}$
Half scattering angle → θ

X-ray Diffraction



Fiber pattern of PE

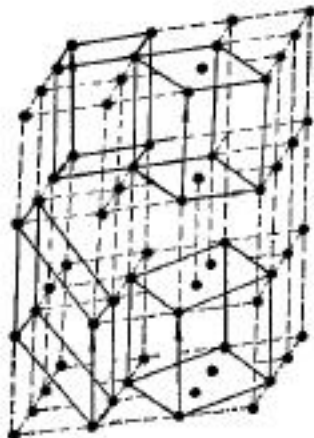
Transmission Electron Microscope Electron Diffraction (TEM ED)



ED pattern of PE

combination

The Seven Crystal Systems

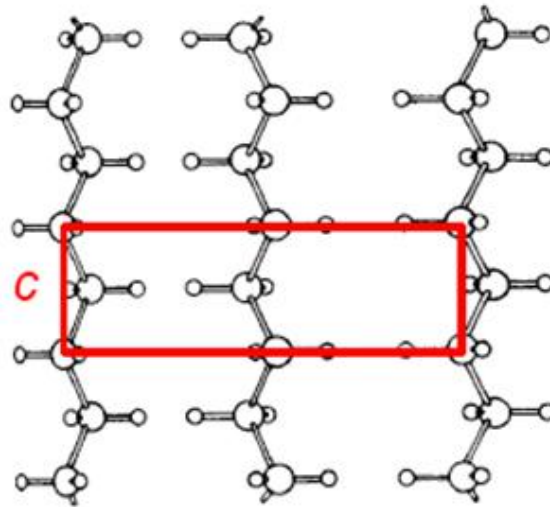


Crystal System	Number of Independent Parameters	Parameters	Lattice Symmetry
Triclinic	6	$a \neq b \neq c; \alpha \neq \beta \neq \gamma$	$\bar{1}$
Monoclinic	4	$a \neq b \neq c; \alpha = \gamma = 90^\circ; \beta > 90^\circ$	$2/m$
Orthorhombic	3	$a \neq b \neq c; \alpha = \beta = \gamma = 90^\circ$	mmm
Tetragonal	2	$a = b \neq c; \alpha = \beta = \gamma = 90^\circ$	$4/mmm$
Trigonal			
rhombohedral lattice	2	$a = b = c; \alpha = \beta = \gamma \neq 90^\circ$	$\bar{3}m$
hexagonal lattice	2	$a = b \neq c; \alpha = \beta = 90^\circ; \gamma = 120^\circ$	$6/mmm$
Hexagonal	2	$a = b \neq c; \alpha = \beta = 90^\circ; \gamma = 120^\circ$	$6/mmm$
Cubic	1	$a = b = c; \alpha = \beta = \gamma = 90^\circ$	$m\bar{3}m$

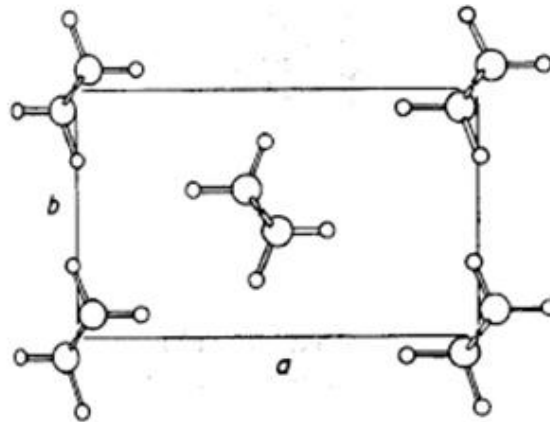
Rare to get single crystals: Powder XRD or films

Polyethylene's Orthorhombic Unit cell

Zig-zag
conformation



$$\begin{aligned} a &= 7.4 \text{ \AA} \\ b &= 4.94 \text{ \AA} \\ c &= 2.55 \text{ \AA} \\ \alpha &= \beta = \gamma = 90^\circ \end{aligned}$$



Vinyl Polymer Crystals

Substituents favor helical conformation

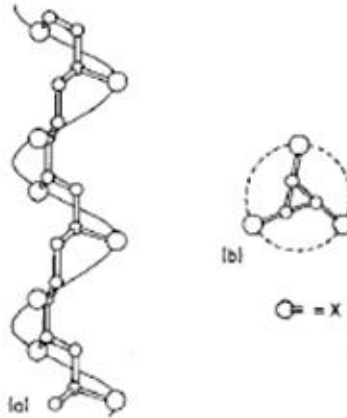


Fig. 4.6 Schematic representation of the $3/1$ helix found for isotactic vinyl polymers of the type $(-\text{CH}_2-\text{CHX}-)_n$. (a) Side view of helix. (b) View along helix. The X groups are $-\text{CH}_3$, $-\text{C}_6\text{H}_5$, $-\text{CH}=\text{CH}_2$ etc.

Helix nomenclature: A^*u/t

A – number of atoms in the main chain

u – number of repeating units in the one repeat of helix

t – number of turns of the helix in one repeat

Ex. PTFE Form II: $1^*15/7$, one C in the main chain, 15 CF_2 turn 7 times in one helix

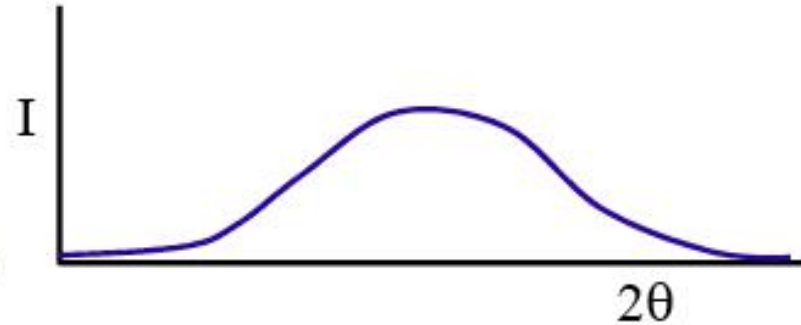
i-PP α -form: $2^*3/1$, two C in the main chain, 3 $\text{CH}_2\text{CH}(\text{CH}_3)$ turn once in one helix

Characterization of Crystallinity in Polymers

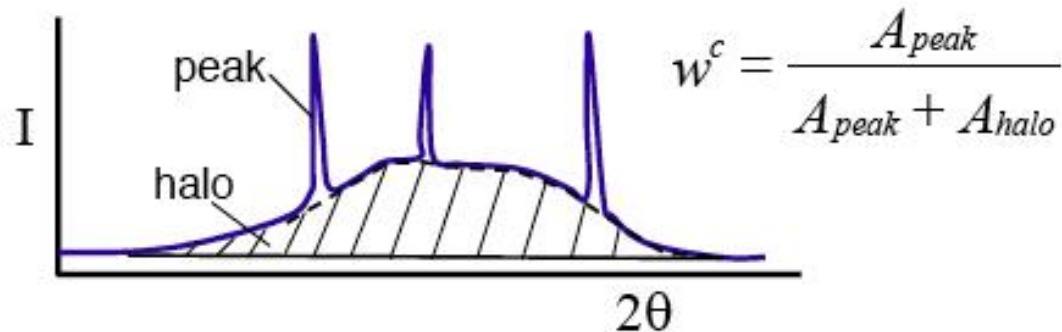
Small-molecule crystals



Amorphous Glassy Polymer



Semicrystalline Polymer



Polymers generally have crystalline and amorphous contributions

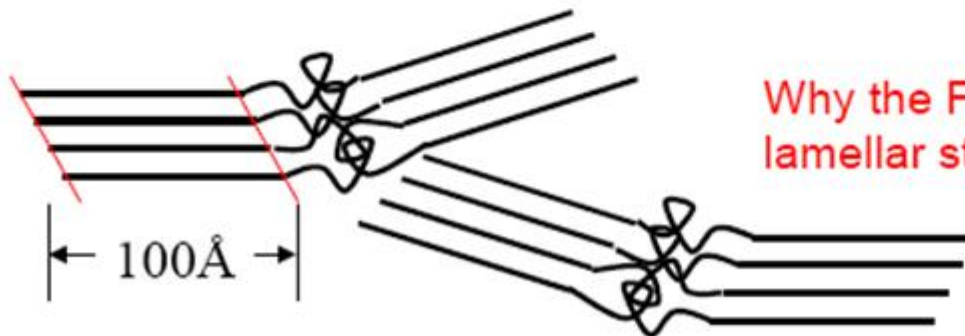
Lamellar Structure of Polymer Crystals

Historical View:

As early as 1928, H. Mark made observations on the structure of crystallites based on the broadening of x-ray diffraction peaks.

Z. Krist. 69 271 (1928).

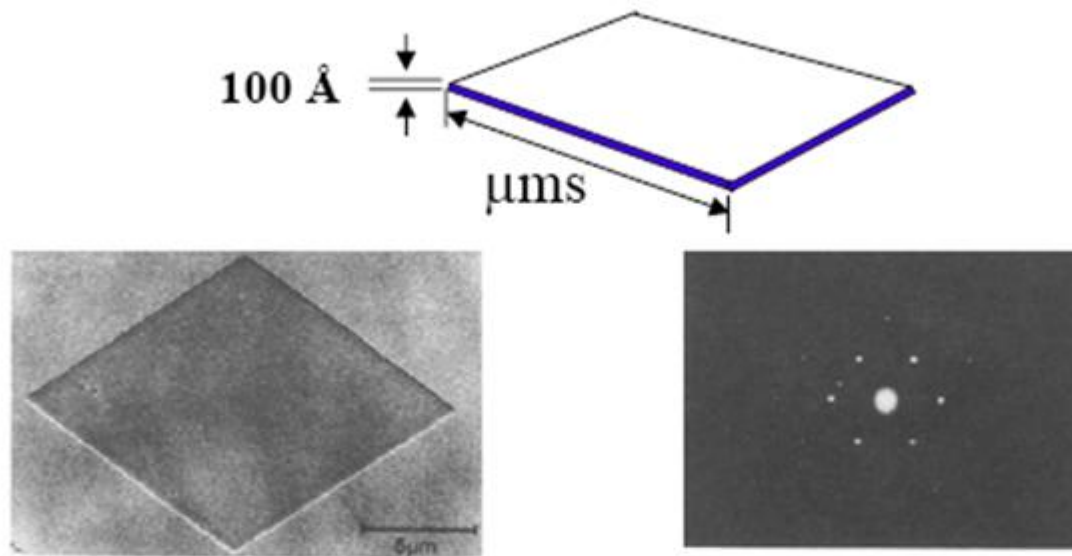
- Estimated that crystallites in native cellulose were about 55 Å wide and 600 Å long.
- Polymer chains are longer than either scale.
- **Fringed-Micelle Model** is proposed.



Why the FM model is a tilted lamellar structure?

Polymer Single Crystals

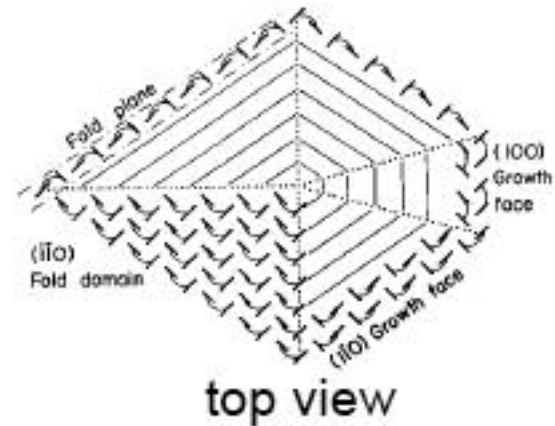
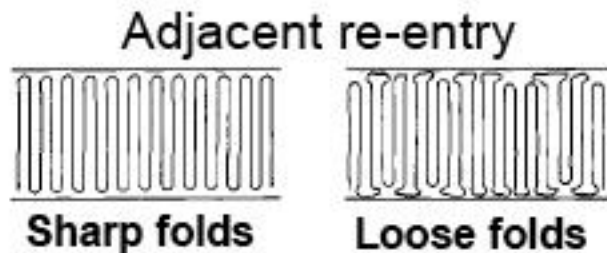
In 1957, three groups (Keller, Till, and Fisher) reported the preparation of single crystals of PE from dilute solution (essentially 100% crystalline PE).



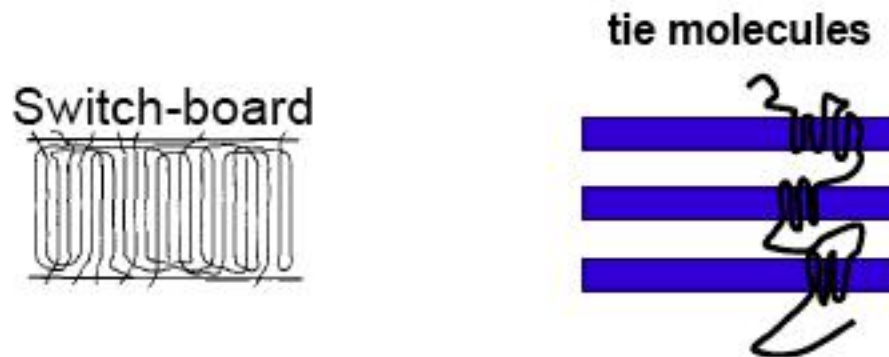
Also, *electron diffraction* data suggested that the chain axes were normal to the face of the single crystals.

Keller $\longrightarrow$ **Chain Folding Model**

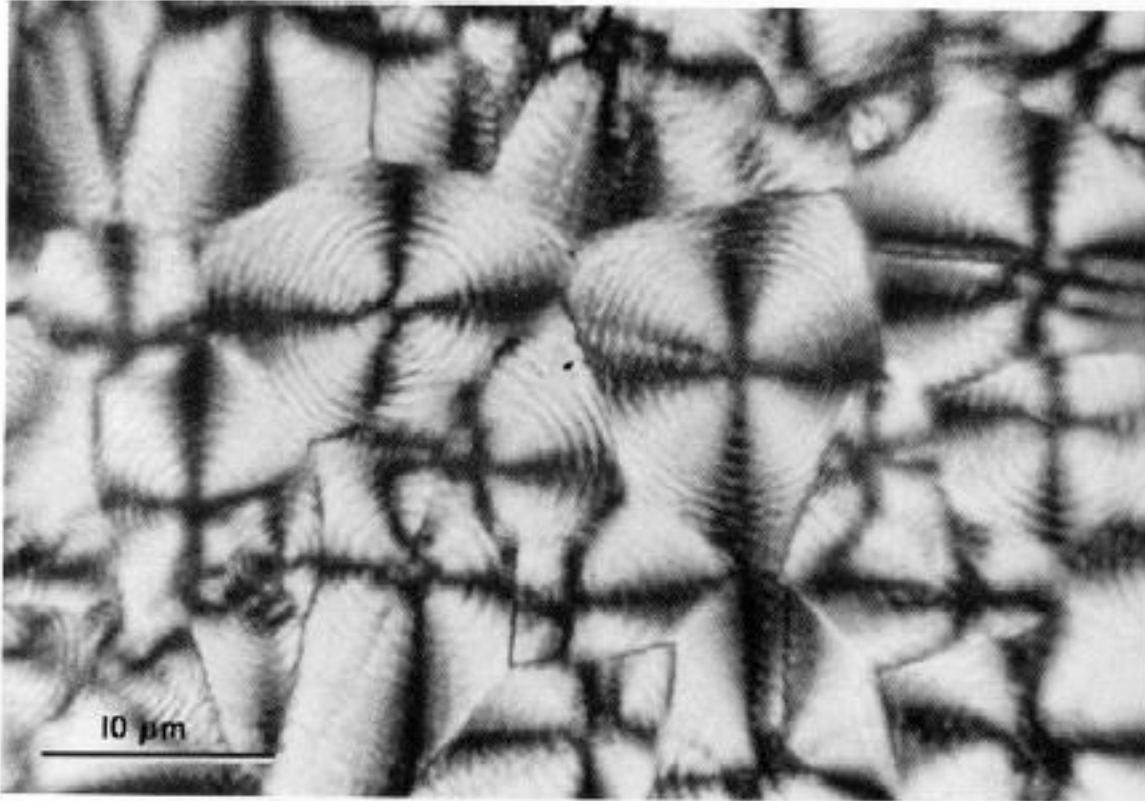
Adjacent Re-entry Model for Solution-grown Single Crystals



Random Re-entry Model for Melt-grown Single Crystals



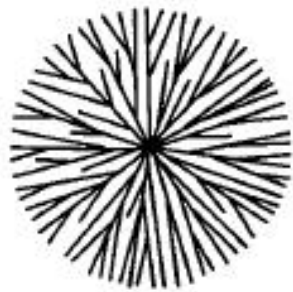
Polyethylene Spherulites



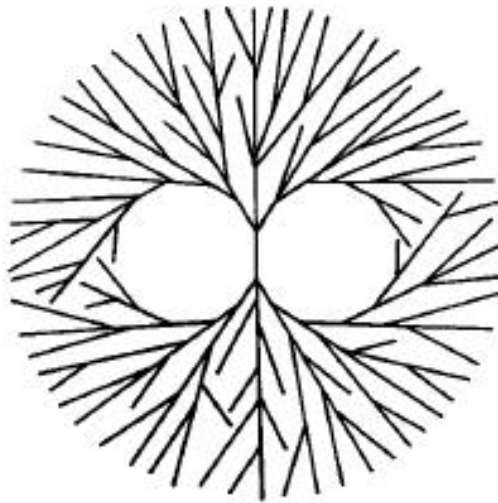
Typical for spherulites are the Maltese crosses which indicate the orientation of polarizer and analyzer and the ring patterns

Crystallized on cooling from the melt, optical micrograph, crossed polarizers.

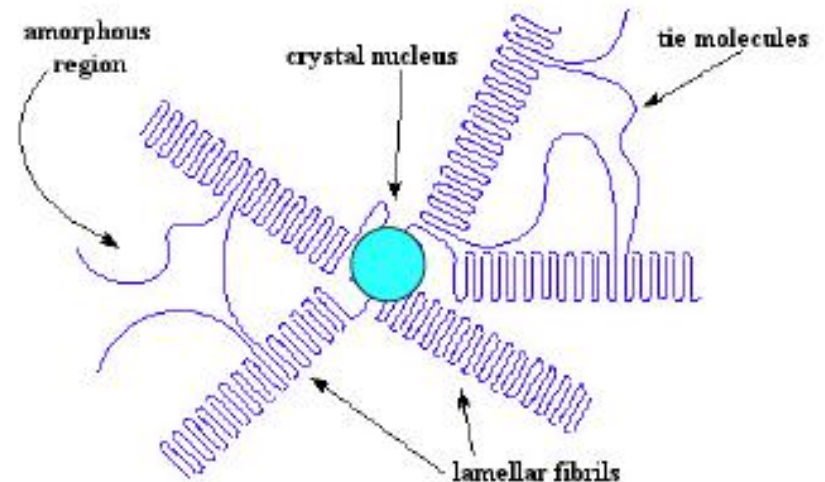
Spherulite Growth from Lamellar Crystals



Hedgehog



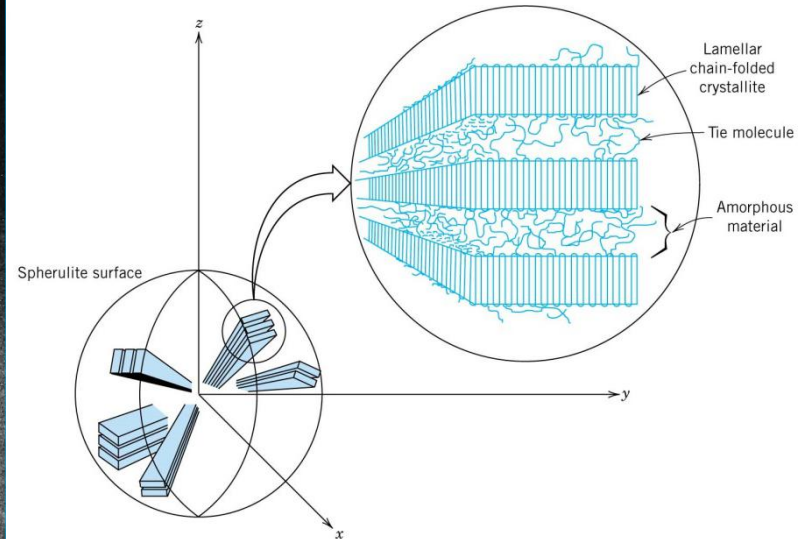
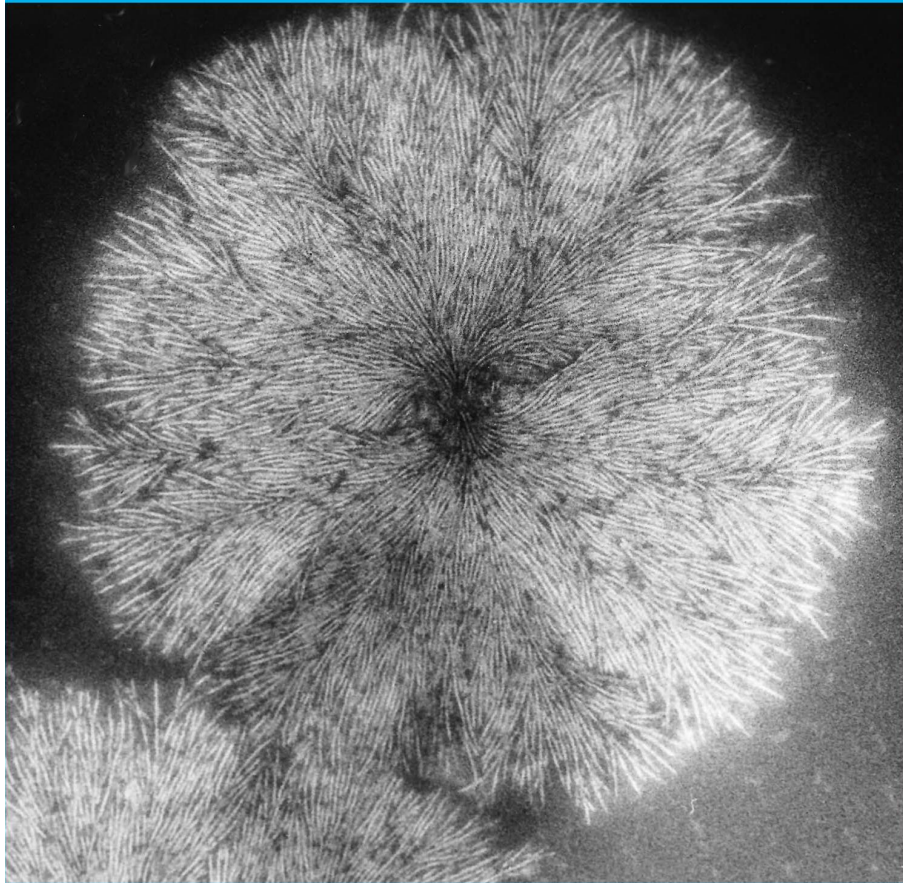
Popoff



a polymer crystalline spherulite

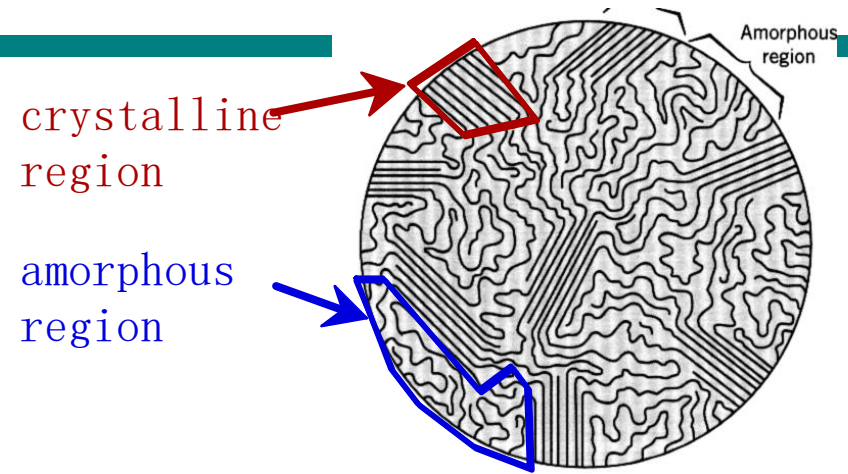
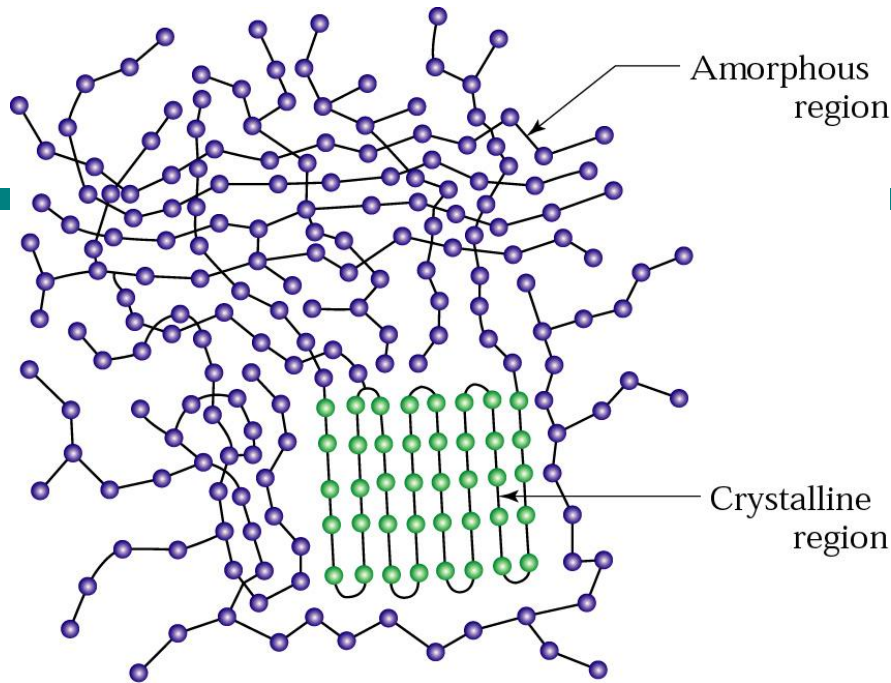
The subcrystal habits maybe fibrillar, lath-shaped or possibly even lamellar.

Crystalline Structures in Polymers



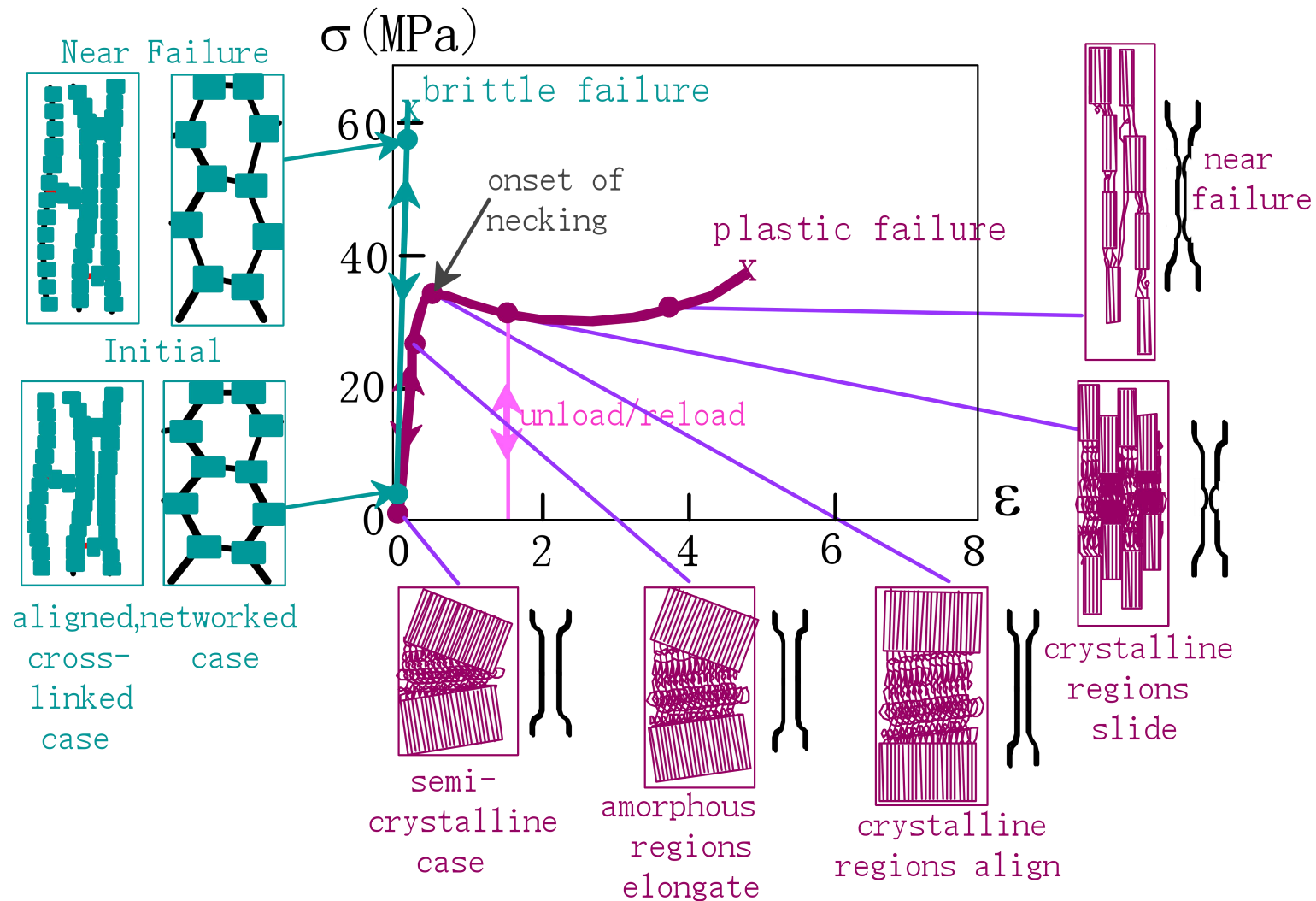
TEM of spherulite structure in natural rubber(x30,000).

- *Chain-folded lamellar crystallites* (white lines) ~10nm thick extend radially.



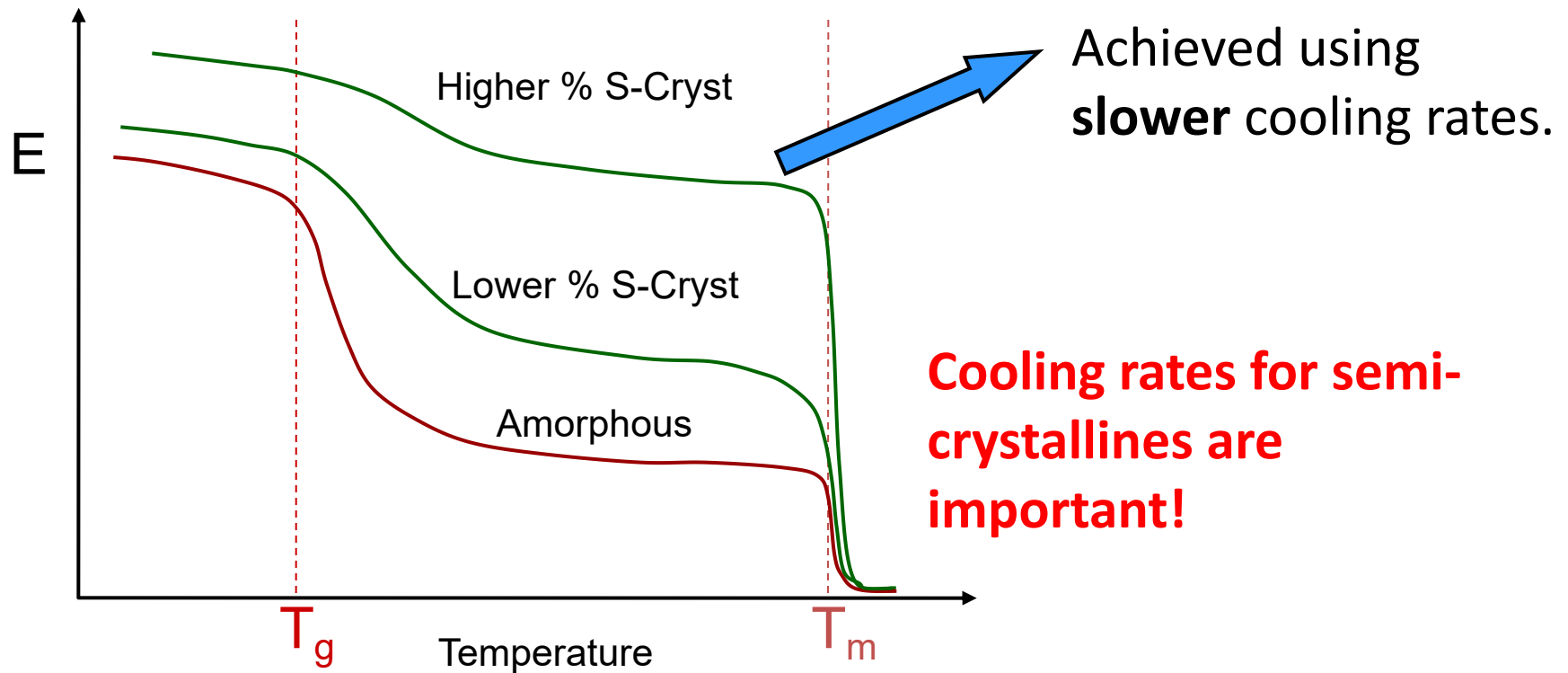
- **Crystallinity:** % of material that is crystalline.
 - TS and E often increase with % crystallinity.
 - Annealing causes crystalline regions to grow and crystallinity increases.

Tensile Response: Brittle & Plastic

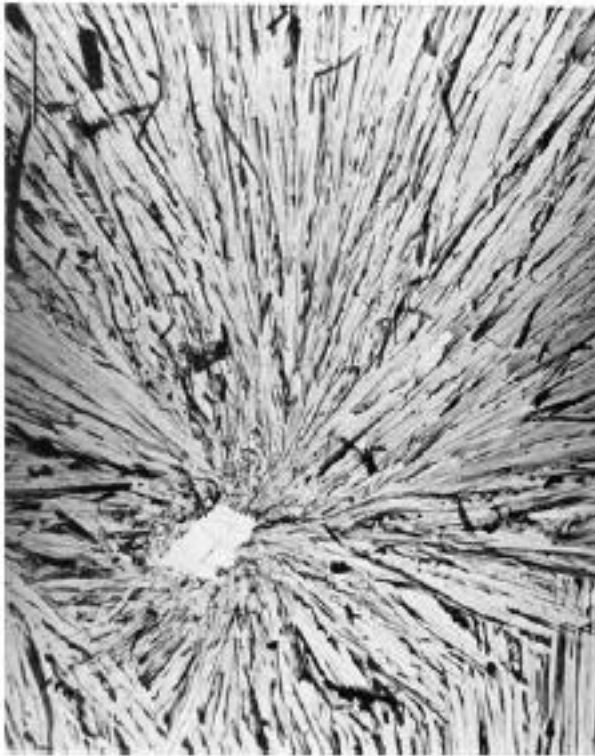


Amorphous polymer properties do not depend on cooling rate.

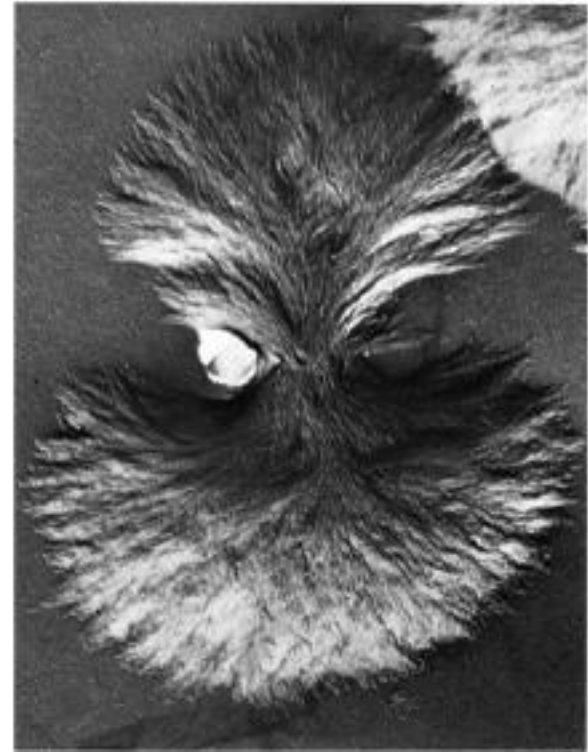
Semicrystalline polymer properties depend on final degree of crystallinity, and hence the rate of cooling.



Micrographs of Polymer Spherulites



A



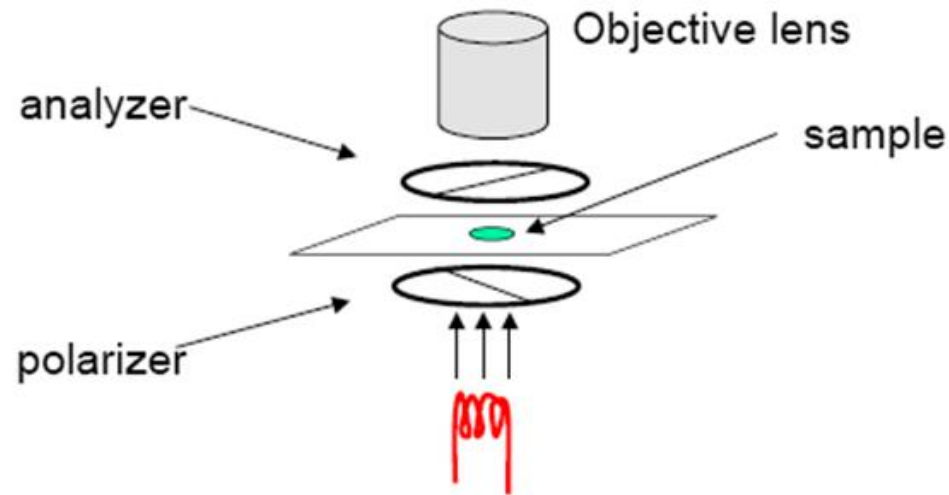
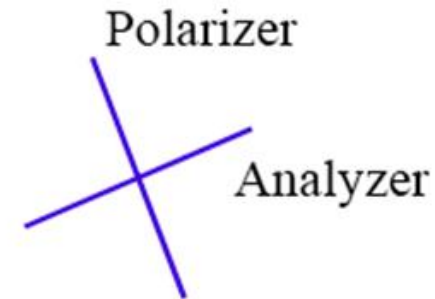
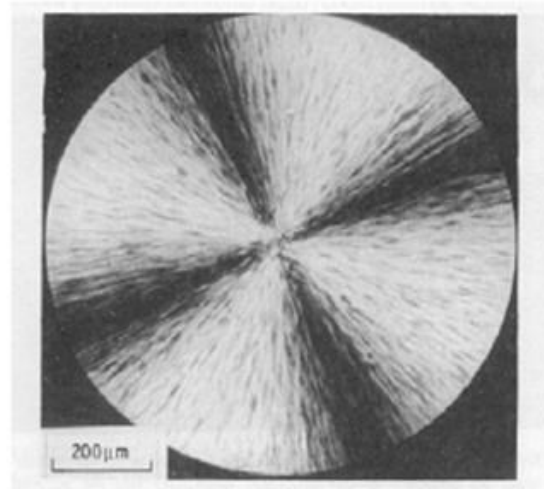
B

1 μm

- A. Fibrous spherulite of selenium (hedgehog type)*
- B. Lath-like spherulite of nylon 6 (popoff-type)*

Seeing **Maltese Crosses**: Polarizing Microscopy

Maltese crosses



Polarizing Optical Microscopy

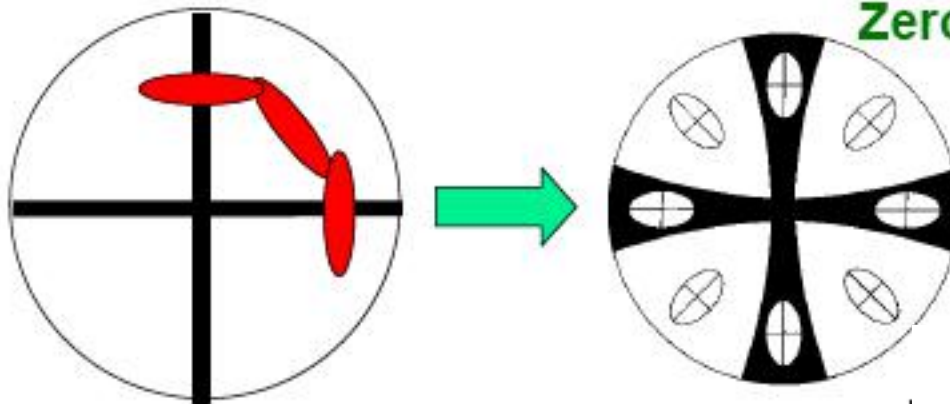
Light transmitted between crossed polarizers:

$$\frac{I}{I_0} = \sin^2(2\alpha) \sin^2\left(\frac{\delta}{2}\right)$$

Angle between birefringence axis and polarizer.

$$\delta = \frac{2\pi L}{\lambda} (n_1 - n_2)$$

Positions within the spherulite where birefringence axis is parallel or perpendicular to the polarizer appear dark. This is where $2\alpha = 0, \pi/2, \dots$



Zero amplitude effect

Formation of Ring Pattern: Lamellar Twisting

Poly(propylene terephthalate)



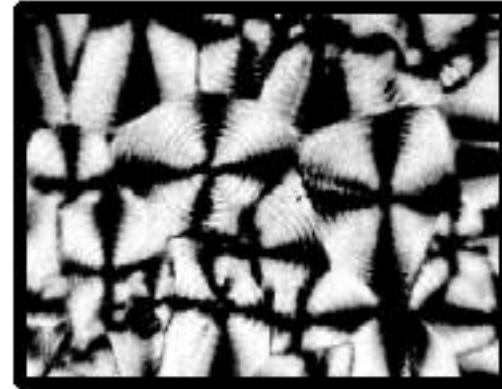
155°C



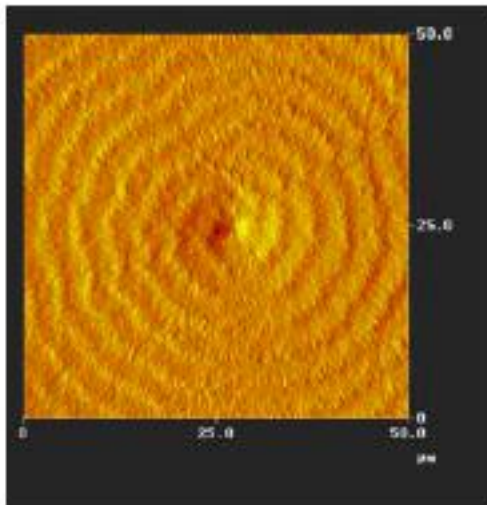
160°C

POM

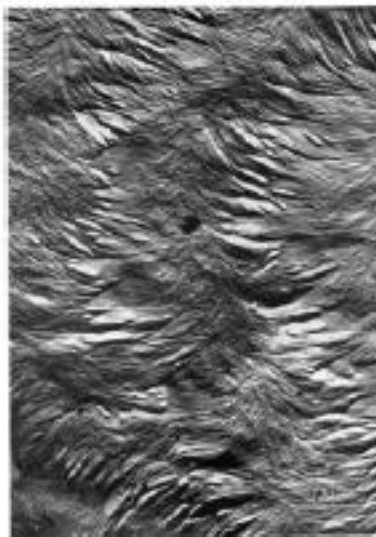
Polyethylene



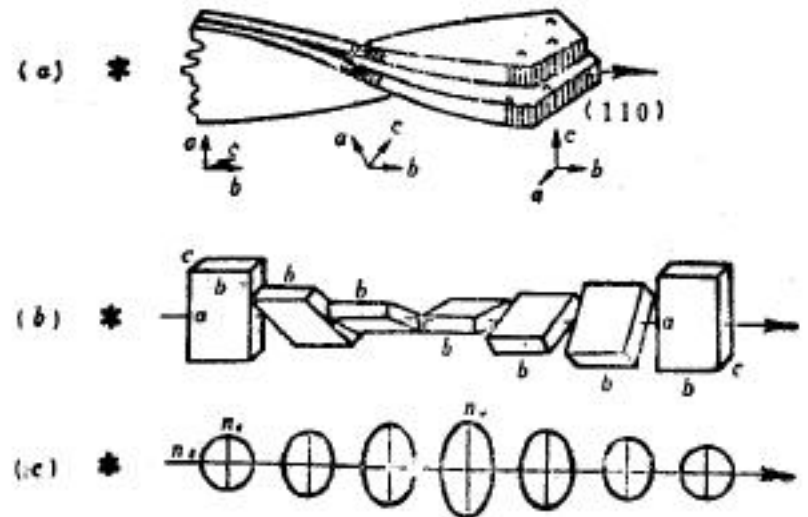
POM



AFM

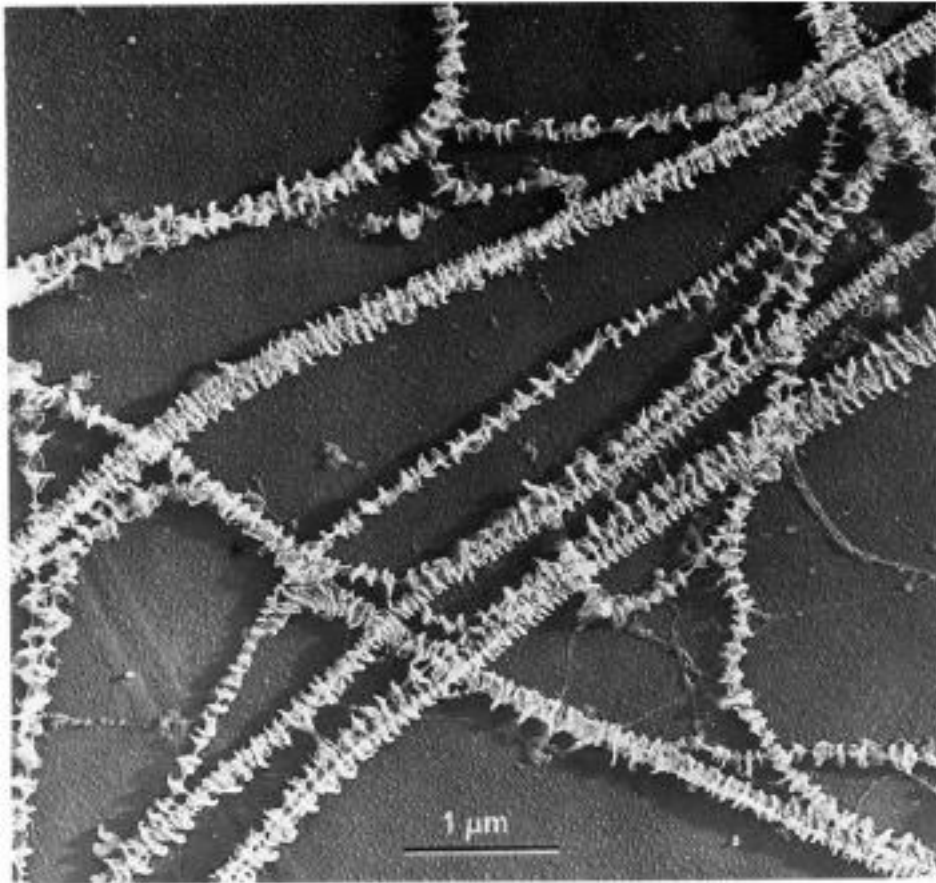


TEM



Polyethylene Fibers Nucleated on Si-C fibers

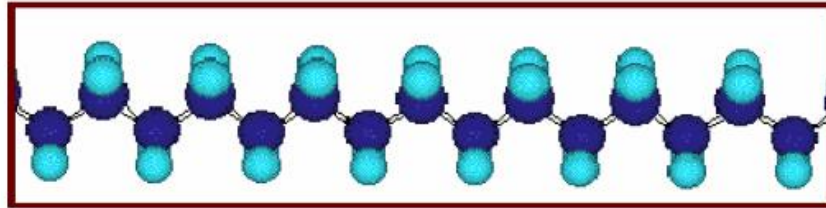
Shish-Kebobs



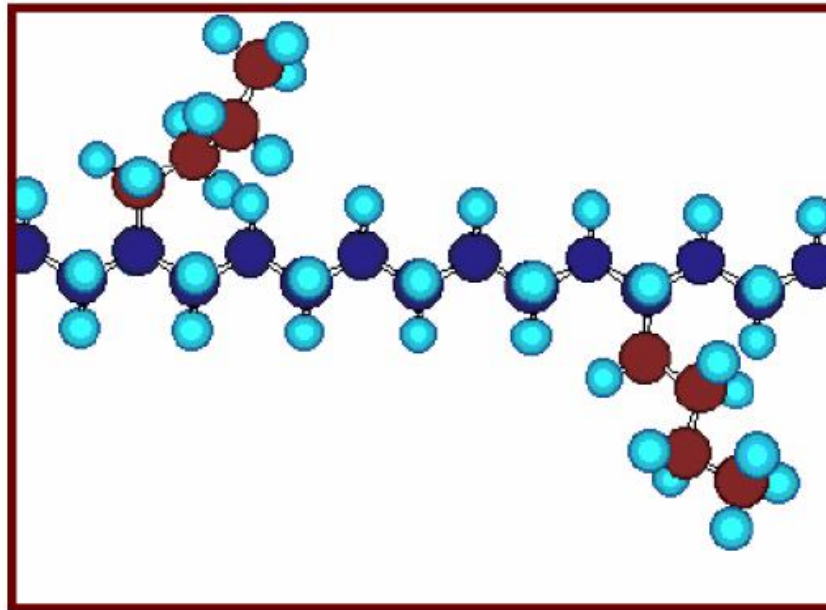
Turbulent or extensive flow is needed for growth of the fibers. From 5% solution in o-xylene, no crystallization above 96 °C in quiet solution, but up to 112 °C crystallization in turbulent flow. (shish-kebobs with decreasing amount of overgrowth)

Branching on Crystallinity

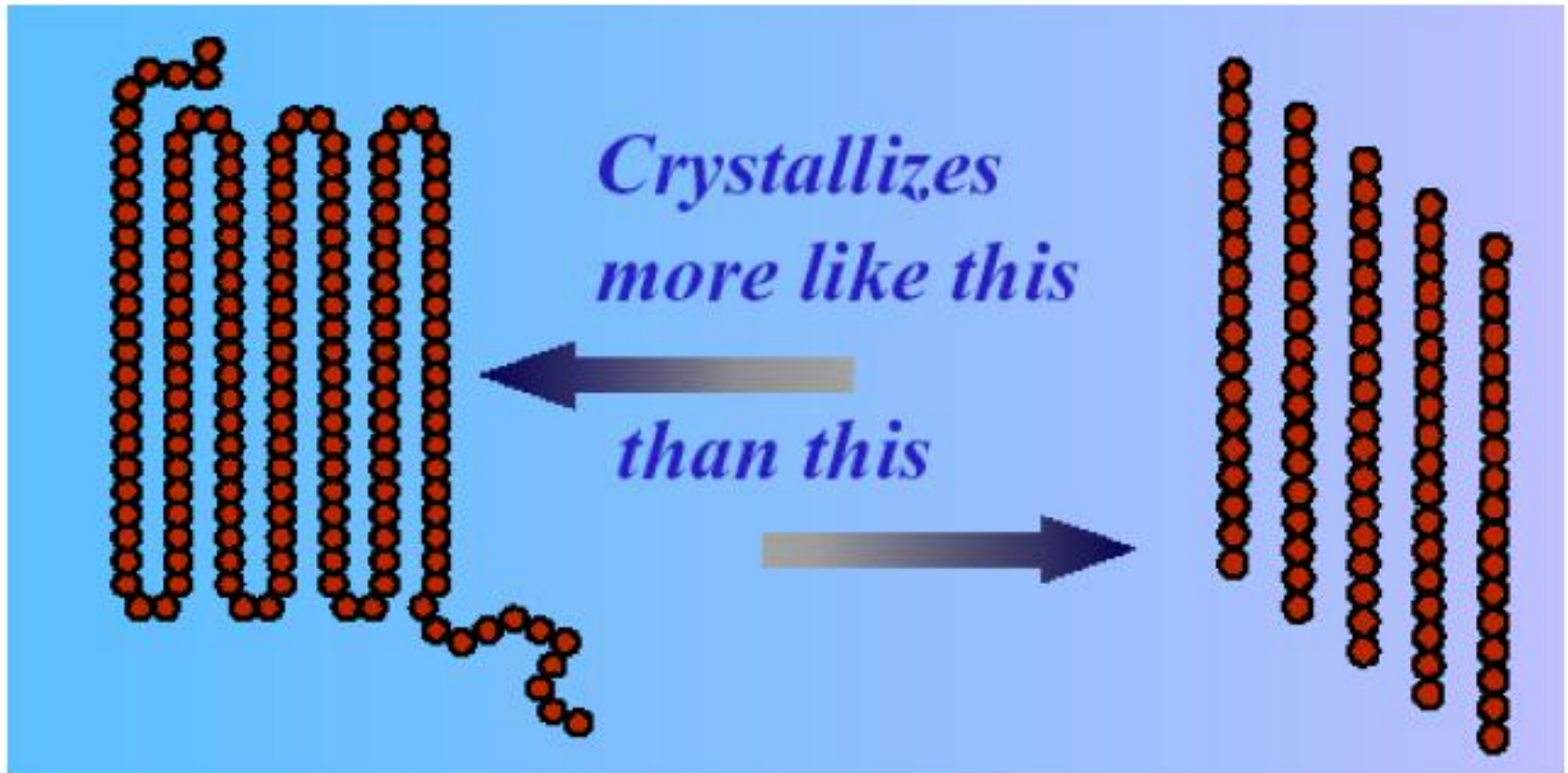
Linear



Branched



Which one will be more likely to crystallize?

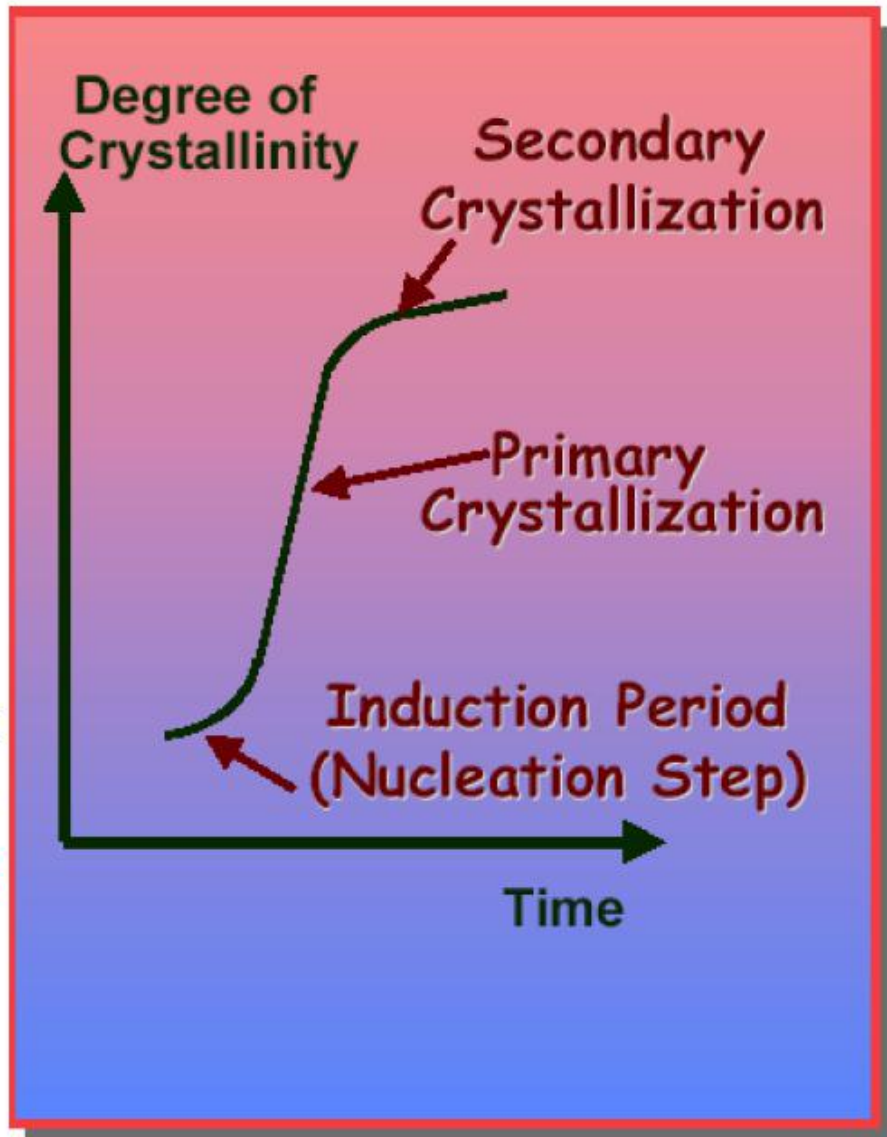
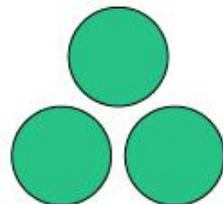


Linear crystallizes easier
(HDPE = linear; LDPE = branched)

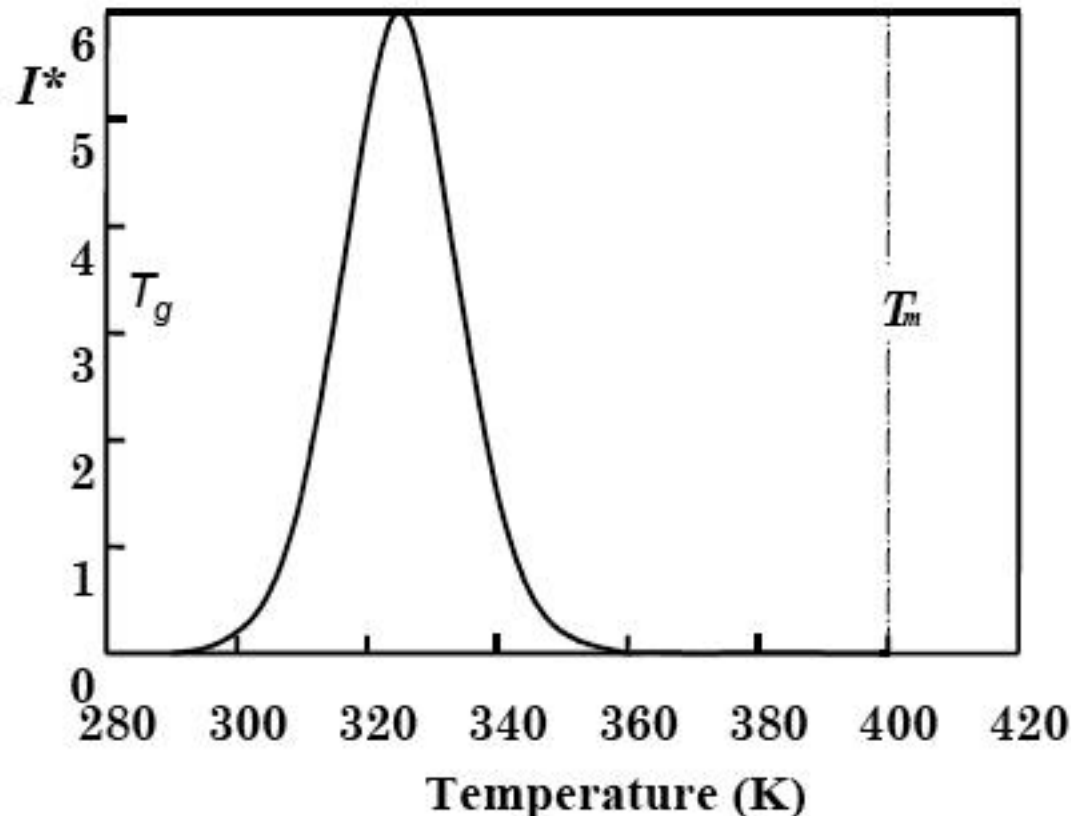
Crystallization Kinetics

General Features

- **Induction period** - Formation of **primary nuclei**
- **Primary crystallization** - A period of fast **spherulitic growth**
- **Secondary crystallization** - A period of slower crystallization that occurs once the spherulites have **impinged on one another**



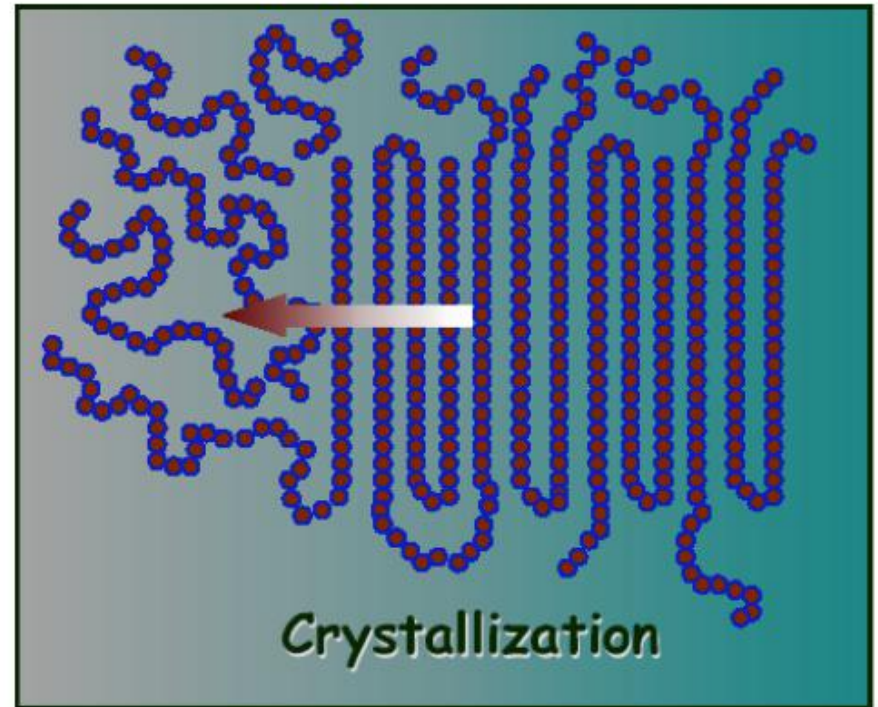
Nucleation Rates between T_g and T_m

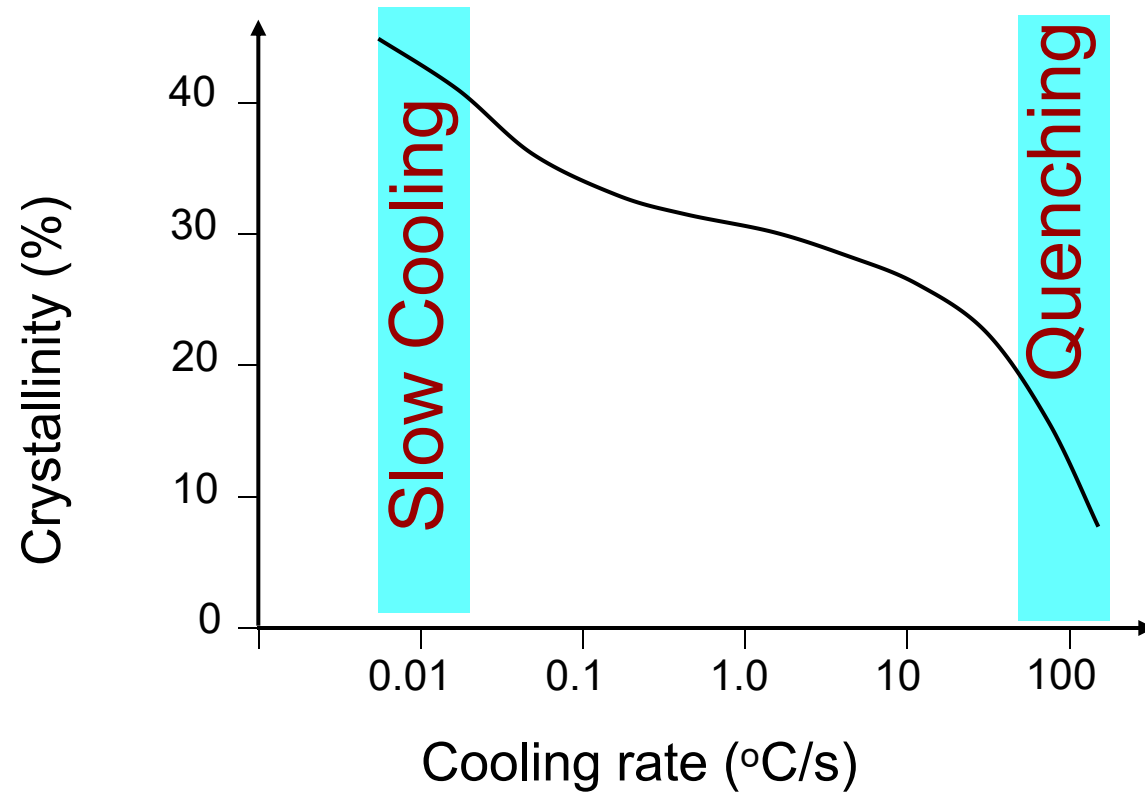


Between T_g and T_m , crystallization takes place. When the T_c is close to T_m , The I^ decreases, while the T_c also decreases with T approaches T_g . This leads to a bell-shaped curve between I^* and T . There is a maximum between T_m and T_g .*

Primary Crystallization

- Once a **nucleus** has been formed **growth** is predominantly in the **lateral direction**.
- There is a considerable **increase** in the **fold period** behind the lamellar front during crystallization from the **Melt**.





Effects of Crystallinity

- 1) **Strength: Stronger & Stiffer**
- 2) **Optical: Opaque (scattering by spherulites)**
- 3) **Higher density**
- 4) **Less Soluble**
- 5) **Less Permeable**
 - **Smaller interchain distances**
 - **Stronger intermolecular forces**

HOMework

- Describe, in detail, how to determine the crystallinity of semicrystalline polymers using DSC and XRD.
- Describe, at least, **3** ways to make polymer crystals.